Decoupling Compression and Routing in Attention for Time-Series Forecasting

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Abstract

Long-horizon time-series forecasting with Transformers is dominated by the cost of quadratic attention over long lookback windows. Compressed-and-indexed attention—compressing the key/value (KV) stream and routing each query to a few compressed blocks—reduces this cost, but its design space is rarely studied systematically. We treat the family as a two-axis design problem: the KV compression rate c_{KV} and the routing resolution c_R. Under a frozen direct-96-step forecasting protocol with twenty-five paired seeds, we run a parameter-equalized 2×2 factorial on ETTh1 and a three-way comparison—full attention, the equal-rate CSA configuration $(4, 4)$ (the design point of DeepSeek-V4's Compressed Sparse Attention), and the dual-rate corner DRRA $(8, 4)$—across twelve datasets spanning the ETT family, large multivariate series, different dynamics, and mechanistic synthetic probes. We report four findings. First, on ETTh1 *every* compressed configuration outperforms full causal attention on validation MSE (mean deltas -18.8% to -27.0%, in 95/100 paired seed comparisons; Holm-adjusted $p \leq 3.3 \times 10^{-6}$) at an equal or slightly smaller parameter count; a bit-exact wiring audit confirms the baseline is correctly implemented, and the advantage survives a projection-matched dense retrain. Second, coarsening the router (c_R:4:1 $\rightarrow$ 8:1) systematically deteriorates accuracy ($+0.030$ validation MSE), while coarsening KV compression is mildly beneficial (-0.032): the pre-registered decoupling hypothesis is supported on ETTh1 (per-seed paired contrast, exact Wilcoxon $p = 0.0074$)—and the conditional reading localizes it, since the interaction is carried by the $(4, 8)$ cell (fine content, coarse router: $+0.0662$) and at coarse content the finer router adds nothing. Routing here is non-learned (the hard top-k selector gives the indexer no task gradient, though route membership still follows the drift of upstream features), so this is a selection-architecture effect, not learned routing. Third, the dual-rate corner DRRA $(8, 4)$ reduces the attention matrix to 48 (+1 sink) columns versus 256 (5.22$\times$) at a matching parameter count. Fourth, the advantage is *signal-dependent*: 8/12 datasets each (DRRA and CSA), concentrated on smooth, redundant real-world series, with slight losses on the synthetic exact-recall probes. The factorial conclusions are statistically established at n=25 on ETTh1 (a three-dataset replication did not reproduce the sign structure); the cross-dataset pattern remains descriptive (five seeds per dataset, per-dataset tests capped at $p = 0.0625$), and latency is reported for the development machine only.

1 Introduction

Long-horizon time-series forecasting—predicting the next H steps from a lookback window of length L—is a core task in energy, traffic, and finance applications. Transformer-based forecasting models (Zhou et al., 2021; Nie et al., 2023; Liu et al., 2024) are accurate but pay a quadratic cost $O(L^2)$ in the attention mechanism as the lookback grows. One widely used remedy is *compressed-and-indexed* attention: compress the L key/value (KV) positions into $C = L/c_{KV}$ blocks, and have each query attend to only a small, *routed* subset of those blocks plus a local sliding window. This family underlies recent efficient-attention designs for large language models (Yuan et al., 2025; DeepSeek-AI, 2026), and its components—content-based routing (Roy et al., 2021), low-rank KV compression (DeepSeek-AI et al., 2024)—are individually well known.

The efficiency stakes are concrete. In large-scale ML, arithmetic work is now the dominant driver of the energy and carbon cost of training and inference (Strubell et al., 2019; Patterson et al., 2021; Schwartz et al., 2020; Wu et al., 2022), and energy is, to a first order, proportional to that work. Dense attention sits on the steepest part of this curve: its dominant term is quadratic in the lookback, so each doubling of L quadruples the attention-layer arithmetic—and, at a fixed hardware generation, the energy per window— until a routing or compression remedy is applied. Long-horizon forecasting turns this from an abstraction into an operating constraint: its deployments (grid and building energy management, traffic control, financial risk systems) are served continuously at sustained query volume, often on power-constrained edge hardware, so per-forecast energy is a real cost rather than an afterthought. For this family, efficiency is therefore not a "nice to have": it is the difference between a forecasting system that can and one that cannot be deployed at the volumes real applications demand. The design question this paper measures—*which compression*

axis buys efficiency, and which axis, when coarsened, is the accuracy we silently give away—is thus also an energy question, and its answer determines how cheaply accurate long-horizon forecasting can be run. We return to that question quantitatively in Section 4.8, where the audited component accounting is used to forecast attention work at contexts up to 2^{17} tokens and to project energy savings under explicit first-order assumptions.

What is not well studied is the *design space* of this family: how the two independent compression axes interact. We identify two axes:

- **KV compression rate** c_{KV}: how aggressively the content stream is compressed into blocks ($C = L/c_{KV}$ blocks);
- **Routing resolution** c_R: how finely the router represents the history when deciding which blocks to attend to ($M = L/c_R$ index tokens).

Coarsening either axis cuts compute, but the two axes may not trade accuracy equally. Our central question is therefore a controlled, empirical one: *which axis carries accuracy, and which axis only buys efficiency?*

This paper is a **design study**, not a proposal of a new attention mechanism: every building block we use is a configuration of known ideas (Section 3), and overlaps with concurrent work (Yuan et al., 2025; DeepSeek-AI, 2026) are acknowledged rather than elided (Section 2). Our contributions are:

(a) **A controlled factorial design with exploratory extensions.** All four combinations of $c_{KV}, c_R \in \{4{:}1, 8{:}1\}$ are evaluated on a frozen task—ETTh1, direct 96-step forecasting—under an *auxiliary-identical* contract (identical window, top-k, heads, hidden size, optimizer, schedule) with attention-parameter equalization to within 0.3%, across 25 paired seeds (Section 4); the three-way comparison (full attention vs. the equal-rate and dual-rate cells) extends to twelve datasets (Section 4.4), and two deeper-ratio probes, $(16, 4)$ and $(32, 4)$, are reported separately as exploratory (Section 4.2, Appendix C).

(b) **A decoupled-rate finding.** Coarsening the router deteriorates accuracy ($+0.030$ validation MSE for c_R:4:1 $\rightarrow$ 8:1), while coarsening KV compression is mildly beneficial (-0.032); the deterioration is concentrated in the $(4, 8)$ cell (fine content, coarse router), so the finding is stated conditionally rather than as a general rule about routing. The pre-registered decoupling hypothesis—the routing effect exceeds the KV effect on per-seed paired differences—is supported at n=25 (exact Wilcoxon $p = 0.0074$). The indexer receives no task gradient (hard top-k is non-differentiable), so the effect is architectural, not learned; route membership still follows the drift of upstream features. The dual-rate corner (c_{KV}=8, c_R=4) (***DRRA***, Section 3), reduces the attention matrix 5.22$\times$ relative to full attention at equal parameters; its edge over the equal-rate cell is not significant on validation (exact Wilcoxon $p = 0.62$).

(c) **A measured efficiency/accuracy characterization.** A surprising—and wiring-verified—baseline result: compressed configurations beat full causal attention on the frozen task (mean deltas -18.8% to -27.0%; Holm-adjusted $p \leq 3.3 \times 10^{-6}$ at n=25), and a projection-matched dense retrain does not close that gap, with explicit limitations (CPU-scale models; across twelve datasets the advantage is signal-dependent, Section 4.4; external architectures are reference points, not peers, Section 4.5).

(d) **Reproducibility artifacts.** A frozen task specification, machine-readable results manifests, a verified parameter registry, causality-verification suites, and a codebase in which every reported number traces to a file (Appendix E).

Scope and non-claims. This work does not claim a novel attention mechanism, does not claim state-of-the-art forecasting accuracy (the external architectures we ran are reference points for calibrating absolute error levels, not peer methods; Section 4.5), and does not claim statistical equivalence. Wall-clock latency *is* reported—as development-machine measurements with a directional-only stance (Table 6, with paired and forecast-required companions in Section 4.7); cross-hardware transfer remains future work. Analytic MACs are likewise reported with explicit caveats (the wrapper MLP dominates; MACs are not energy). Pre-freeze latency figures from earlier stages of this project are superseded and are deliberately *not* used here.

2 Related Work

2.1 Routed and compressed attention

The general idea of content-based routing goes back to the Routing Transformer (Roy et al., 2021), which clusters tokens with online k-means and restricts attention to tokens in the same cluster. Native Sparse Attention (NSA) (Yuan

et al., 2025) combines coarse-grained token compression with fine-grained token selection, making it the closest conceptual cousin of the family studied here. DeepSeek-V4's hybrid attention (DeepSeek-AI, 2026) pairs a Compressed Sparse Attention (CSA) stage with a Heavily Compressed Attention (HCA) stage for million-token contexts, reporting 27% of single-token inference FLOPs and 10% of KV cache versus DeepSeek-V3.2; we adopt the working name "Dual-Rate Routed Attention" (DRRA) for our family in part to avoid colliding with that label, and refer to the equal-rate configuration point that matches CSA's rate settings as "CSA", following DeepSeek's terminology (Section 3). The low-rank KV bottleneck and shared-query projection used in our configurations descend from Multi-head Latent Attention (MLA) (DeepSeek-AI et al., 2024). Attention distillation toward cheaper inference architectures (Goldstein et al., 2025) overlaps a secondary, separable thread of this project (Appendix B) rather than the routing/compression core studied here.

Crucially for our claim (b), none of these works, to our knowledge, *systematically varies the content-compression rate and the routing resolution independently and reports their separate main effects* on a fixed task. The closest are Simplified Sparse Attention (Mao et al., 2026), which compresses context into gist tokens and selectively unfolds top-k chunks but derives k from the chunk size to hold the total budget constant (Eq. A.1), coupling the two axes; and The Sparse Frontier (Nawrot et al., 2025), which sweeps selection budgets across six methods with no content-compression knob. DeepSeek-V4 (DeepSeek-AI, 2026) and NSA (Yuan et al., 2025) expose both mechanisms as fixed hyperparameters without varying them independently. Two structural differences are worth naming. First, terminology: DeepSeek-V4 has no "router" — its *Lightning Indexer* scores compressed KV entries and a *top-k selector* subsamples them (Section 3.3 adopts the generic term "router" for this scorer-plus-selector pair). Second, in CSA the indexer's key stream is compressed with the *same* operation and rate m as the KV entries, so the scoring granularity is tied to the KV compression rate; in our family the indexer stream has its own rate c_R, which is precisely the axis the factorial varies. We confirmed this on a systematic sweep of the 2024–2026 routing/compression-attention literature (August 2026).

Mixture-of-block attention (MoBA) (Lu et al., 2025) is the closest selection-only relative: it partitions the raw KV cache into blocks and routes each query to the top-k blocks through a gating function, so block size sets the selection granularity and k the realized budget. The contrast sharpens both axes of our design. First, MoBA's content stream is the uncompressed cache — the only granularity knob is the block size, and it is the same knob as the selection unit; our family instead compresses the content stream at rate c_{KV} (KV-cache compression) and scores a separately compressed index stream at rate c_R (routing resolution), which is what lets the two axes vary independently. Second, holding the top-k count fixed while c_R changes — our operating protocol — changes how many valid blocks each query actually reads, whereas MoBA reports block size and k jointly and discusses holding the attended budget roughly constant across granularities; the realized-coverage caveat in Section 4.3 states the consequence for our fixed-k comparison, with a budget-matched control deferred rather than assumed away (Appendix F). Third, MoBA trains its selection pathway end to end, so its query and key streams adapt to the routing they induce, while our indexer is parameter-free and receives no task gradient (Section 4.3 measures what that implies for route stability in practice).

2.2 Efficient and accurate time-series forecasting

Long-horizon forecasting architectures include the ProbSparse-attention Informer (Zhou et al., 2021), patching-based PatchTST (Nie et al., 2023), the inverted iTransformer (Liu et al., 2024), linear baselines (Zeng et al., 2023), and 2D-variation modeling with TimesNet (Wu et al., 2023). Two of them, DLinear (Zeng et al., 2023) and PatchTST (Nie et al., 2023), have been run under our frozen protocol as external reference points (not peer methods; Section 4.5); the rest are itemized as pending work in the appendix future-work list, Appendix F. Our contribution is orthogonal: we do not propose a new forecasting architecture but study two design axes of the attention module that any of these models could use.

2.3 Positioning summary

This paper sits in the intersection of efficient attention and time-series forecasting, as a *controlled design study*: it measures how two configurable axes of a fixed attention family trade off accuracy and efficiency on one frozen task, and it does so with the reproducibility discipline (pre-registered factorial, paired seeds, parameter equalization, frozen task spec) that makes the trade-off structure measurable rather than anecdotal.

3 Method

3.1 Task and evaluation protocol (frozen)

All experiments use a single frozen task specification—stated in full in this section and frozen before any model-comparison run—applied to twelve datasets: the ETT family (ETTh1, ETTh2, ETTm1, ETTm2; 7 features, hourly and 15-minute) (Zhou et al., 2021), the large multivariate sets Electricity (321 channels), Traffic (862), and Weather (21) and the Exchange rates (8, daily), as benchmarked by Wu et al. (2021) and redistributed by the Time-Series Library (Wu et al., 2023), Solar-Energy (137, 10-minute) (Lai et al., 2018), and three mechanistic synthetic families (controlled sinusoids, change points, and a delayed-copy task; 7 channels each) generated deterministically from fixed seeds (Appendix F). Each dataset is split chronologically $60\%/20\%/20\%$ into train/validation/test with stride-1 sliding windows of length $L{=}256$ that never span a split boundary. Normalization is a per-feature z-score whose statistics are fit on the *train* split only and applied to all splits (the earlier practice of fitting over the full series was a future-statistics leak and is fixed); channels with zero training variance are normalized with unit scale. The task is *direct* 96-step forecasting: in one forward pass the model maps $x \in \mathbb{R}^{B\times 256\times F}$ to $\hat{y} \in \mathbb{R}^{B\times 96\times F}$ (F features per dataset), and the loss is the pooled MSE over all $(B, 96, F)$ elements, reported in standardized units (lower is better). No autoregressive decoding and no teacher forcing is used. Causality is enforced by construction and verified empirically: a future-invariance and exact-zero-future-gradient suite passes for every variant. Windows overlap by 255/256 positions and are therefore not independent samples; this is a declared limitation (Section 5.3).

Training is identical for every configuration: AdamW ($\mathrm{lr}{=}10^{-4}$, weight decay 10^{-5}), gradient clipping at norm 10, batch size 32, a fixed budget of 10 epochs, float32 on CPU (Apple Silicon, 8 threads), and deterministic seeding with the same seed used for data loading and model initialization. The factorial is run at $n{=}25$ paired seeds: the five pre-registered seeds $\{42, 123, 2024, 7, 99\}$ and a 20-seed power extension (seeds 1–21 excluding the pre-registered set) run under the byte-identical protocol and merged post hoc (Section 4.2). The fixed 10-epoch budget is justified by learning-curve analysis on the pre-registered-seed runs: all cells are in a plateau regime after epoch 4 and the mean last-3-epoch drift across those 20 runs is $+0.0009$; no cell keeps improving systematically, so the budget does not bias cell comparisons.

3.2 Model family

All configurations share one minimal Transformer block (hidden size 128, 4 heads, head dimension 32) with a horizon head $\mathrm{Linear}(128 \to 96 \cdot F)$ applied to the last hidden state:

$$x = \mathrm{input_proj}(u) + \mathrm{pos_embedding}, \qquad h = \mathrm{MLP}(\mathrm{LN}(\mathrm{attn}(x) + x)), \qquad \hat{y} = \mathrm{horizon_head}(h[:, -1, :]), \quad (1)$$

The only difference between configurations is the attention module. The shared Q/KV/O projections follow the MLA-style pattern: a low-rank query projection with RMSNorm, a tied single-head KV projection (K = V) with RoPE applied to the last 8 of 32 dimensions, and a grouped O projection. A full causal attention variant (*Normal*) with no compression serves as the task-appropriate baseline.

3.3 The compressed-and-indexed attention family

Each compressed configuration (Figure 1) consists of three components, all inherited verbatim from the reference implementation:

Two-series compressor. The L positions are projected to double width and gated by a per-dimension softmax over $2m$ slots, so that every token contributes to two adjacent windows (its own and the next); the weighted sum is RMSNormed and RoPE-encoded at block starts. With rate m, this yields $C = L/m$ KV blocks covering $[mj, mj + m - 1]$. The same compressor, with its own parameters, produces the $M = L/m_{idx}$ routing index tokens.

Router (scorer + top-k selector). Each query scores all M index tokens ($\mathrm{ReLU}(q \cdot \mathrm{idx}^\top)/\sqrt{d}$, head-mixed by a weight that is fixed in practice — see the trainability note below), masks tokens that lie in the future, and selects the top-k ($k{=}16$). We use "router" for the scorer-plus-selector combination; DeepSeek-V4 (DeepSeek-AI, 2026) calls the analogous scorer the *Lightning Indexer* and the selection step the *top-k selector* (their paper uses neither "router" nor a standalone indexer module feeding the selector — the indexer scores the KV entries, which the selector then subsamples). In all compressed configurations the router reuses the main query (shared-Q), valid because the query head dimension equals the index head dimension; this removes one projection and one RoPE application. The hard top-k selection is non-differentiable, so the indexer parameters (including CSA's dedicated $\mathrm{idx_q_proj}$) receive zero task-loss gradient: no learning signal reaches the selector, and "routing granularity" effects in this paper are selection-architecture effects, not learned routing. The released audit logs both a zero gradient norm and a zero optimizer

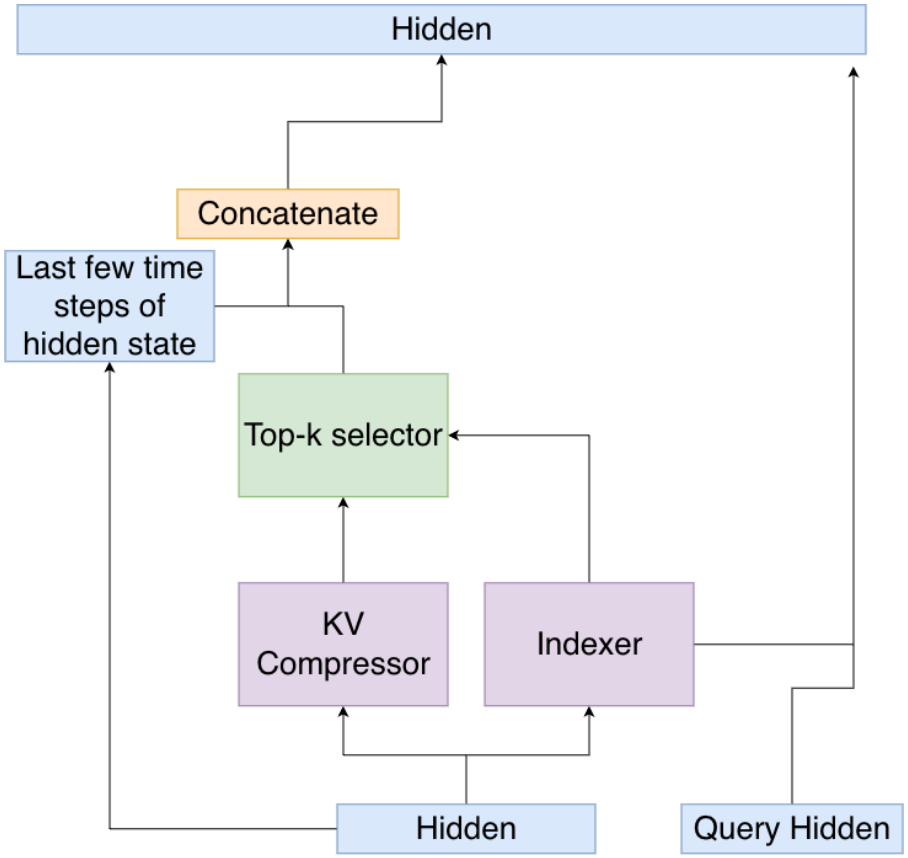


Figure 1: Architecture of the compressed-and-indexed family (the $(c_{KV}, c_R) = (8, 4)$ cell drawn; c_{KV} = KV-cache compression rate, positions per stored block; c_R = routing resolution, positions per index token). One shared Transformer block whose only varying component is the attention module: a KV compressor (c_{KV}=8: 256 positions $\rightarrow$ 32 blocks), a routing indexer (c_R=4: 64 index tokens) with top-k=16 selection (the router keeps the 16 best-scoring index tokens per query), and dense masked attention over the fixed column set (32 KV blocks + 16-token sliding window + 1 always-visible sink column $= 49$ columns; Normal = full attention, attends to all 256). The horizon head maps the last hidden state to the 96-step forecast.

update delta for every indexer-exclusive parameter, so those parameters are tracked by the optimizer rather than absent from the graph. The audit establishes gradient isolation, not literal route constancy—index scores are computed from features produced by trainable upstream components, so the selected set can change as those features drift over training. A route-overlap probe across epochs measures that drift rather than assuming it away (Section 4.3); a trained selector remains deferred future work (both audit limits are in the released logs).

Dense masked attention over the sparse selection. Instead of gathering the selected blocks, attention is computed densely over a fixed column set—the C KV blocks plus a sliding window of W=16—with $-\infty$ masking of unselected blocks and a single learnable "sink" softmax column for stability. Masking to $-\infty$ makes the dense result exactly equal to the sparse top-k selection, with no approximation. On CPU this formulation is several times faster than per-query gather in our implementation, and it keeps the softmax exact.

Index-to-block mapping (per cell). The router selects top-k=16 index tokens; each token t covers $[4t, 4t{+}3]$ at c_R=4 ($[8t, 8t{+}7]$ at c_R=8) and must be translated to KV-block ids before the dense masked attention:

- $(8, 4)$: two fine tokens per coarse block — block $j = t \mathbin{/\!/} 2$, covering $[8j, 8j{+}7]$ (at most two of the 16 selections collide per block).
- Equal-rate cells $(4, 4)$, $(8, 8)$: token t *is* block t (identity).
- $(4, 8)$: one coarse token fans out to two fine blocks $(2t, 2t{+}1)$, doubling the per-query selection to 32 block

slots.

- Deeper ratios ($(16, 4)$, $(32, 4)$): p fine tokens per coarse block — block $j = t \mathbin{/\!/} p$ with $p = 4, 8$ (the general many-to-one rule).

A selected block is read only when its entire span ends at or before the query ($\text{block_end} \leq q$); otherwise — and for tokens masked as future or out of range — it maps to a sentinel id that _attend drops, so no query reads future positions. Causality holds through the overlapping compressor as well: each compressed block mixes only its own and the previous window (receptive field $[m(j-1), mj+m-1]$), i.e. strictly backward, so the block-end guard is sufficient. Any other (c_{KV}, c_R) geometry raises instead of guessing a mapping (the general rule above covers every tested cell). *Historical note:* before 2026-09-28 the implementation applied $t \mathbin{/\!/} 2$ unconditionally, misrouting every cell but $(8, 4)$; the audit, fix, and re-run of the affected cells are logged in the source release.

3.4 The two design axes and the factorial

The two axes are the KV compression rate $c_{KV} = m$ and the routing resolution $c_R = m_{idx}$. We evaluate the 2×2 factorial over $c_{KV}, c_R \in \{4{:}1, 8{:}1\}$, i.e. cells $(4, 4)$, $(8, 4)$, $(4, 8)$, $(8, 8)$. The two corners that also appear in the cross-dataset study are referred to by name throughout:

- **CSA** — the equal-rate corner $(4, 4)$, in which both axes sit at $4{:}1$. We adopt the *name* of DeepSeek-V4's Compressed Sparse Attention (DeepSeek-AI, 2026), whose rate design point this cell matches: KV compression at rate $m{=}4$ (the same value used in DeepSeek-V4-Flash/Pro), routing/selection over entries at the same $4{:}1$ granularity, and a local sliding-window branch. The name marks the configuration, not an equivalence claim: we evaluate our own implementation at $L{=}256$ scale (selection budget $k{=}16$, window 16, versus their $k{\in}\{512, 1024\}$, window 128), not DeepSeek-V4's mechanism (Section 2).

- **DRRA (Dual-Rate Routed Attention)** — the dual-rate corner $(8, 4)$ (coarse content, fine routing), which gives the family its working name. We use "champion" for this cell strictly as shorthand for "the dual-rate corner" (its efficiency geometry, not accuracy rank: on validation no cell is best, DRRA-vs.-CSA exact Wilcoxon $p = 0.62$); everywhere else we name the cell explicitly.

The remaining corners $(4, 8)$ and $(8, 8)$ appear only in the factorial.

Why only $\{4, 8\}^2$? The grid is the pre-registered design (frozen before any run), and a 2×2 factorial is the minimal design that identifies both main effects and their interaction — exactly the three fixed effects of the deferred mixed-effects model. Grid extension was a deliberate exclusion, for three reasons. First, the natural next points are degenerate rather than informative: at $c_R{=}16$ the indexer stream has only $L/16 = 16$ tokens, so the top-$k{=}16$ router selects *all* of them and routing collapses to attend-to-everything — a $(\cdot, 16)$ cell would measure the no-routing endpoint, not finer routing (measuring the latter would require raising k as well, breaking the auxiliary-identical contract that pairs all cells). Symmetrically, at $c_{KV}{=}16$ the KV blocks (16) no longer outnumber the sliding window (16), so the column composition changes what is being measured. Second, the two levels bracket a $2\times$ range around the operating region of interest (the CSA corner $(4, 4)$ matches DeepSeek-V4's $m{=}4$), and both effect signs are established within the bracket. Behavior outside the bracket was extrapolation at pre-registration time; two deeper ratios, $(16, 4)$ and $(32, 4)$, were run later as exploratory probes (same protocol) and are reported separately from the confirmatory factorial (Section 4.2, Appendix C). The grid remains valid as the confirmatory design: the probes stay exploratory and never enter the confirmatory contrasts. Third, budget: each additional cell costs 25 paired runs ($\approx$ 6 h CPU at the measured per-run cost of the seed extension), and the 2×2 spends that budget on replication — statistical power for the pre-registered contrasts — rather than on extra cells. Adding off-grid cells (e.g. the $(\cdot, 16)$ endpoint ablation, roughly 1–2 h for 5 paired seeds) is straightforward in the harness and is itemized as future work. Table 1 lists the resulting geometry and parameter counts.

Reading the table. Normal is the uncompressed reference: it attends to all 256 columns and carries 102,496 attention parameters. Every compressed cell shares the identical auxiliary contract (window 16, top-k 16, 4 heads, hidden 128, shared-Q, ranks 64) and differs only in the two rates. $c_{KV}{=}8$ halves the number of KV blocks (32 vs. 64) and thereby the softmax columns ($48{+}1$ vs. $80{+}1$; 256/49 complete columns, i.e. $5.22\times$ fewer than Normal); c_R sets how many routing candidates the router selects from (64 for $c_R{=}4$, 32 for $c_R{=}8$) without changing the attention cost. All compressed cells carry slightly *fewer* attention parameters than Normal (-6.7% to -6.9%; the low-rank *q/o* bottlenecks replace dense projections), and full-model counts are $\sim 2\%$ below Normal's. The DRRA $(8, 4)$ cell therefore combines the smallest attention geometry among the factorial cells with the fine-router candidate count.

Table 1: Configuration geometry and parameters (measured; attention module and full model). All compressed cells share the auxiliary contract (window 16 = tokens visible to every query, top-k 16 = best-scoring index tokens routed per query, 4 heads, head dimension 32, hidden 128, shared-Q = one query projection shared by router and attention, *q*/*o* ranks 64 = low-rank bottleneck dimensions) and differ only in (c_{KV}, c_R) = (KV-cache compression rate, routing resolution, in positions per stored block and per index token); the low-rank bottlenecks are equalized to within 0.27% of the $(8, 4)$ reference; results for all cells are in Table 2. Starred rows are post-hoc exploratory probes, excluded from all confirmatory contrasts. Columns: C = KV blocks, M = index tokens, *softmax cols* $= C + W + 1$ = KV blocks + 16-token window + always-visible sink column. Normal attends to all 256 columns with no sink.

Config	c_{KV}	c_R	KV blocks C	Index tokens M	Softmax cols	Params (attn / full)
Normal	—	—	—	—	256	102,496 / 355,200
$(4, 4)$	4	4	64	64	80 + 1	95,396 / 348,100
$(8, 4)$	8	4	32	64	48 + 1	95,652 / 348,356
$(4, 8)$	4	8	64	32	80 + 1	95,652 / 348,356
$(8, 8)$	8	8	32	32	48 + 1	95,908 / 348,612
$(16, 4)^*$	16	4	16	64	32 + 1	96,164 / 348,868
$(32, 4)^*$	32	4	8	64	24 + 1	97,188 / 349,892

Realized routing is sparser than the dense-masked budget (about 14 of 64 unique blocks for equal-rate cells, about 10 of 32 for the dual-rate cell; compute still pays the dense-masked width).

Parameter equalization. The factorial is pre-registered with an auxiliary-identical contract: all cells share window, top-k, heads, hidden size, optimizer, and schedule, and differ only in (c_{KV}, c_R)–the cell definition. Because the position-bias terms of the two compressors grow with their rates, attention-parameter counts differ slightly between cells; the only free compute-quality knobs not part of the cell definition (the low-rank q and o bottlenecks) are fixed at the DRRA $(8, 4)$ ranks $(64, 64)$, which is the lattice minimum for *every* factorial cell. The largest remaining attention-parameter delta among the four factorial cells is 0.268% (full-model delta $\leq 0.074\%$), well inside the pre-registered 0.5% tolerance; the exploratory cells keep the same ranks with disclosed deltas ($+0.54\%$, $+1.6\%$).

Complexity. Attention cost per query is proportional to the number of softmax columns: 256 for Normal, 80 for c_{KV}=4 cells, 48 for c_{KV}=8 cells (5.22× fewer complete columns than Normal), and 32/16 for the exploratory $(16, 4)$/$(32, 4)$ cells, giving $O(L \cdot \text{cols} \cdot d)$ per sequence for the score/value computation. The router, however, compares every query against $M = L/c_R$ index tokens, $O(L \cdot M) = O(L^2/c_R)$: the total attention-path cost is therefore *quadratic in L with a reduced constant* ($1/c_R$ on the routing term), *not* subquadratic. Routing cost is identical for the two c_R=4 cells regardless of c_{KV}. These are structural facts; measured mean latency, batch scaling, and empirical log-log slopes are reported in Section 4.7 (Tables 6, Figure 7); FLOPs and memory accounting are reported there as well. Cross-hardware transfer and depth/width ablations are conscious scope exclusions of this study (single machine; single-block architecture), left to future work.

4 Experiments

4.1 Setup and data integrity

The factorial comprises 4 compressed cells $\times$ 25 paired seeds plus the full-attention baseline at the same 25 seeds $= 125$ runs under the frozen protocol of Section 3.1. The five pre-registered seeds $\{42, 123, 2024, 7, 99\}$ were run first (audited manifest with parameter-equalization and training-match checks); a 20-seed power extension (seeds 1–21 excluding the pre-registered set) was run later under the byte-identical protocol and merged post hoc (seed list fixed in the extension design before any extension run; the merge was planned at that time). Three cells ($(4, 4)$, $(4, 8)$, $(8, 8)$) were subsequently re-run under a corrected index-to-block mapping (the original applied one mapping rule to all geometries; Section 3.3) and merged by seed, replacing their rows; the $(8, 4)$ and Normal rows are untouched originals. Data-integrity checks on the merged manifest confirm: (i) parameter equalization within tolerance (max attention delta 0.268%, max full-model delta 0.074%); (ii) matched training across all runs (final training MSE span 0.2756–0.3029); and (iii) all 125 runs present, 25 seeds in every cell, no duplicates. A separate 25-run manifest holds the projection-matched dense baseline introduced in Section 4.6; it follows the same protocol and seed list, and no other reported row depends on it. Reported validation and test metrics come from the epoch-10 weights; no best-validation checkpoint selection is used. The 25 seeds capture training randomness on the fixed dataset and protocol,

Table 2: Factorial results (25 paired seeds, frozen direct-96 protocol; pooled MSE (mean squared error) in standardized units, lower is better). Rows are cells (c_{KV}, c_R): c_{KV} is the KV-cache compression rate (positions per stored block) and c_R the routing resolution (positions per index token); the two dense reference rows are Normal (as-run) and Normal-r64 (projection-matched, Section 4.6). "Beats $(4,4)$" counts seeds where the cell's validation MSE is better than the same seed's $(4,4)$ value. Bold: best (lowest) test mean among the four confirmatory cells. Starred rows are post-hoc exploratory probes, excluded from all confirmatory contrasts (Section 4.2).

Cell	Val mean $\pm$ std	Test mean	Train mean	Beats $(4,4)$
Normal	0.8389 ± 0.0814	1.0804	0.2864	1 / 25
Normal-r64	0.7910 ± 0.0649	1.0281	0.2975	1 / 25
$(4,4)$	0.6148 ± 0.0704	0.8771	0.2941	—
$(8,4)$	0.6190 ± 0.0827	**0.8440**	0.2876	16 / 25
$(4,8)$	0.6810 ± 0.0570	0.9496	0.2863	6 / 25
$(8,8)$	0.6128 ± 0.0660	0.8505	0.2837	12 / 25
$(16,4)^*$	0.5810 ± 0.0751	0.8100	0.2793	16 / 25
$(32,4)^*$	0.6122 ± 0.0652	0.8586	0.2702	14 / 25

not future-period or dataset uncertainty.

4.2 RQ1: the 2×2 factorial

Table 2 and Figure 2 summarize validation MSE by cell over 25 paired seeds. No single cell dominates: the best means are $(8,8)$ (0.6128 ± 0.0660), CSA $(4,4)$ (0.6148 ± 0.0704), and DRRA $(8,4)$ (0.6190 ± 0.0827)—all within seed noise of each other—while $(4,8)$ is the worst compressed cell (0.6810 ± 0.0570). Every compressed cell beats the full-attention baseline on mean validation and mean test MSE; per-seed, 22–25 of 25 paired comparisons favor each compressed cell (95/100 in total). Head-to-head on validation against CSA, DRRA wins 16/25 and (8,8) wins 12/25 while (4,8) wins 6/25; best-of-four counts are 8 CSA, 8 (8,8), 7 DRRA and 2 (4,8), so there is no best cell. On test the same DRRA-vs-CSA contrast favors DRRA by -0.0331 (95

Main effects. Averaging over the opposite axis (Figure 3): coarsening KV compression from 4:1 to 8:1 *lowers* validation MSE by -0.0320 (95% CI $[-0.062, -0.002]$; mildly beneficial), while coarsening routing resolution from 4:1 to 8:1 *raises* it by $+0.0300$ (95% CI $[+0.001, +0.059]$; deterioration; Wilcoxon $p = 0.059$, paired t $p = 0.043$). Signs are consistent in 20 of 25 (KV) and 16 of 25 (routing) seed-column comparisons. The interaction is significant: -0.0724 (95% CI $[-0.129, -0.016]$, Wilcoxon $p = 0.014$), and reading it as conditional effects localizes it. Per seed, coarsening KV compression is nil at fine routing ($+0.0042$, 95% CI $[-0.044, +0.052]$, exact Wilcoxon $p = 0.62$, 9/25) and beneficial at coarse routing (-0.0682, $[-0.101, -0.036]$, $p = 0.0001$, 21/25), while coarse routing costs accuracy at fine KV content ($+0.0662$, $[+0.027, +0.106]$, $p = 0.002$, 19/25) and is nil at coarse content (-0.0062, $[-0.048, +0.035]$, $p = 1.0$, 13/25). The interaction is therefore carried by the $(4,8)$ cell: at c_{KV}=8 the finer router does not improve the mean at all, so the two axes are *not* additive and the main effects are not available as general mechanism statements—they must be read conditionally, and the routing main effect in particular is a statement about the fine-content configuration (paired per-seed intervals for all four conditional effects, and for the two main effects and the interaction, are in the released analysis log).

Pre-registered hypothesis H1 asked whether coarsening the router deteriorates performance *more* than coarsening KV compression. At n=25 paired seeds the strict form is *supported*: on per-seed main-effect differences the routing effect exceeds the KV effect by $+0.0620$ on average (exact paired Wilcoxon $p = 0.0074$; paired t-test $p = 0.0107$; full recomputation from unrounded seeds in the released analysis log). This per-seed contrast is one of four reported estimates (the two main effects, the interaction, and the direct DRRA-vs.-equal-rate difference below)—it pairs all four (c_{KV}, c_R) observations within a seed, and it is algebraically identical to the off-diagonal difference $y_{4,8} - y_{8,4}$, so the off-diagonal alone carries no independent information. Both individual effects remain modest relative to seed noise ($\sim$0.03 vs. mean within-cell std 0.070, per-cell range 0.057–0.083; sign consistency 20/25 (KV) and 16/25 (routing)), and — with a significant interaction and no single best cell — we report the decoupling as a *statistically supported sign structure with non-additive axes* whose mechanism-level reading is still interpretation (Section 5; the mixed-effects model MSE $= \beta_0 + \beta_1 \log_2 c_{KV} + \beta_2 \log_2 c_R + \beta_3 \log_2 c_{KV} \log_2 c_R +$ random effects is deferred as a robustness re-expression (the 2×2 with seed blocking already identifies all three fixed effects; more rate levels would add shape, not identification)). Per-seed trajectories show the ordering across cells is not unanimous for every seed—with the top

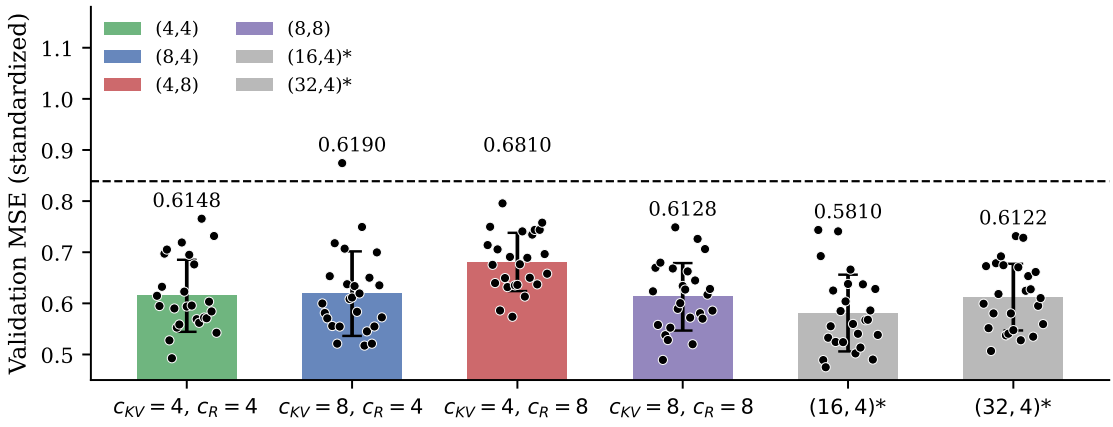


Figure 2: Validation MSE (mean squared error) by factorial cell; bars are means over 25 paired seeds, error bars $\pm$ one seed standard deviation, and each dot is one seed's value. Lower is better. The dashed line marks the full-attention (Normal) baseline at 0.8389 validation MSE: bars below it beat full attention. The $(8, 4)$ corner $(c_{KV}{=}8, c_R{=}4)$ combines coarse KV compression with fine routing. Starred bars (labels with $*$) are post-hoc exploratory probes, excluded from all confirmatory contrasts.

three cells tied within noise, there is no "best" cell to claim at all.

What the c_{KV} contrast changes. The factorial compares complete configurations at a fixed selection budget $k{=}16$: coarsening KV compression also coarsens the span each selection covers, so realized coverage is not held fixed (the measured unique selections are about 14 of 64 blocks for equal-rate cells and about 10 of 32 for the dual-rate cell; Section 3). The KV main effect is therefore a statement about the configuration—content density at a fixed selection count—and not about content density at matched realized coverage. Budget-matched controls that fix the number of unique valid blocks after mapping, and coverage-matched sensitivity analyses across c_{KV}, are the natural next comparisons and are deferred (Appendix F).

Cross-dataset replication of the factorial. The full four-cell factorial was repeated on three contrasting datasets (ETTh2, Electricity, Exchange; five seeds each, same protocol and corrected mapping). Cell means (CSA/DRRA/C48/C88 validation MSE): ETTh2 1.78/1.81/1.69/1.99; Electricity 0.378/0.381/0.387/0.380 (all within 0.009); Exchange 3.14/3.24/3.09/3.09. The ETTh1 sign structure does not travel: on ETTh2 the H1 contrast is -0.1217 (opposite sign from ETTh1; exact Wilcoxon $p = 0.63$), on Exchange it is -0.1446 ($p = 0.81$), and on Electricity all four cells sit within 0.009 of each other (H1 $+0.0058$, $p = 0.0625$ — too small to call). DRRA-vs.-CSA is a tie on all three (W/L 2/3, 2/3, 3/2; all $p \geq 0.6$). The decoupling is therefore an ETTh1 finding, not a general design law (no cell leads best-of-four on any of the three datasets) — consistent with the task-dependent reading of Section 4.4.

Exploratory probe: pushing KV compression to 16:1. Motivated by the flat KV-rate profile above, one further cell, $(16, 4)$ (16 KV blocks, 64 index tokens), was run post hoc on ETTh1 (25 seeds, same protocol; ranks held fixed with a disclosed $+0.5\%$ parameter delta). It does not join the confirmatory contrasts (shown starred for reference), but the outcome bounds the claim from above: $(16, 4)$ reaches 0.5810 ± 0.0751 validation (0.8100 test), the best mean of any cell and ahead of Normal in all 25 seeds. Coarser KV content is not merely free here — it is mildly beneficial, consistent with the negative KV main effect; the knee, if any, lies beyond 16:1. A further exploratory probe at $(32, 4)$ (8 KV blocks) lands at 0.6122 ± 0.0652 validation (0.8586 test), still inside the band (beating Normal 24/25): the profile stays flat out to 32:1, and 4 KV blocks is the sensible floor where this study stops.

4.3 RQ2: compressed vs. full attention

Table 3 compares full causal attention (Normal, 256 columns, no compression) with the two cells that anchor the family's design corners, CSA $(4, 4)$ and DRRA $(8, 4)$, and adds the projection-matched dense reference Normal-r64 (Section 4.6). Every compressed cell beats Normal on mean validation and mean test MSE, and *95 of the 100 paired seed comparisons* favor the compressed configuration (per-seed validation deltas span -48.3% to $+9.9\%$ across cells; mean deltas -18.8% to -27.0%). DRRA $(8, 4)$ beats CSA $(4, 4)$ head-to-head in 16 of 25 seeds on validation (CSA

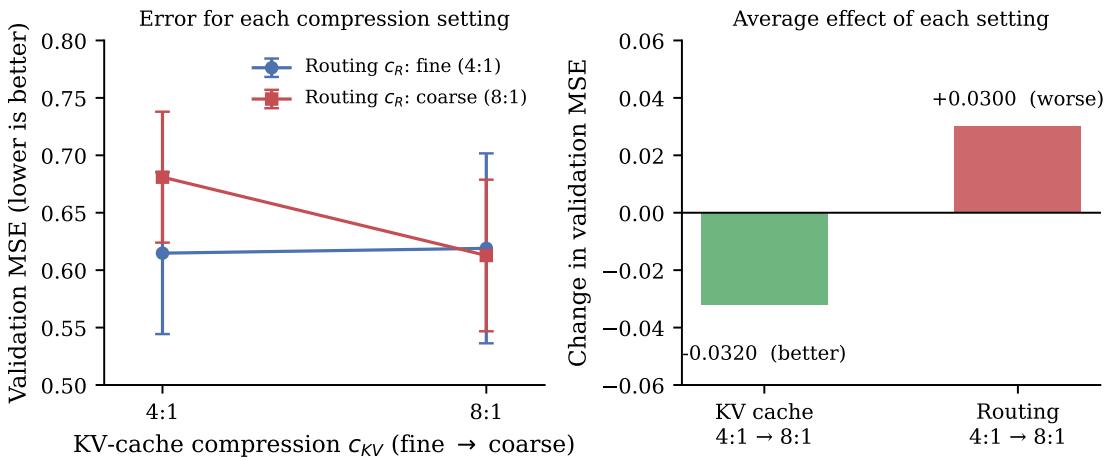


Figure 3: Compressed attention on ETTh1: validation MSE (mean squared error; lower is better) over 25 paired seeds. **Left:** error for each of the four compression settings; lines join the two routing resolutions (fine 4:1, coarse 8:1) as KV-cache compression goes 4:1 → 8:1; error bars are within-cell seed std (mean 0.069). The two settings are not independent: coarsening KV compression lowers the mean only under coarse routing (-0.0682 at c_R=8:1 against $+0.0042$ at c_R=4:1), and the routing penalty is concentrated at fine KV content ($+0.0662$ against -0.0062). **Right:** each setting's effect averaged over the other axis (c_{KV} 4:1 → 8:1: -0.0320, mildly beneficial; c_R 4:1 → 8:1: $+0.0300$, deteriorates). The per-seed paired contrast (routing effect > KV effect) is significant (exact $p = 0.0074$; the interaction is -0.0724, $p = 0.014$, so the axes are not additive).

ahead in 9), and no cell is best of all four in more than 8 (DRRA 7, $(8, 8)$ 8, CSA 8, $(4, 8)$ 2; merged corrected-mapping analysis in the released logs). The advantage does not depend on the dense baseline's wider output projection: retrained with the compressed cells' rank-64 output projection (25 paired seeds, same protocol), the dense model improves to 0.7910 validation MSE but every compressed cell still beats it (paired validation deltas -0.110 to -0.178 MSE, all four 95% intervals excluding zero, 21–25 of 25 seeds per cell; Section 4.6).

Wiring verification. Because the compressed variants beating an unmodified baseline by ∼18–26% is a surprising direction, we audited the Normal attention implementation before reporting it: a from-scratch naive reference (true causal masking, $1/\sqrt{d}$ scaling, independent RoPE reimplementation) matches the module output bit-exactly at L=32 and L=64 (0.0 max absolute difference) and to 1.19×10^{-7} at L=256 (fp32 accumulation noise); future positions receive exactly zero weight and future-token perturbation leaves outputs unchanged. The baseline is correctly implemented, so the comparison stands. We also note the direction is surprising in the other sense: compression acting as a task-appropriate inductive bias (see Section 5) is a plausible but *unverified* explanation *that three mechanistic probes now support descriptively* (models retrained with per-run checkpoints under the same frozen protocol; seed 42; regenerated against the current implementation after the mapping correction): (i) on the delayed-copy dataset (five channels periodic with in-window lags 96–256, plus two noise channels), *all three* models learn the exact-offset copy — per-channel copy MSE 0.0007 (Normal) and 0.0013–0.00318 (compressed cells) against a noise floor of ≈1.05 — so exact-offset recall is not the differentiator, though Normal is ∼2× more precise on the copy channels, with a mild lag gradient for the compressed cells; (ii) on the change-point dataset, CSA $(4, 4)$ has the lowest per-window error in *every* distance-from-change bucket (the across-bucket gradient is confounded by the generator's random-walk drift and should not be over-read); and (iii) across the twelve datasets, the compression advantage correlates with signal smoothness in the predicted direction (Pearson $r = -0.29$ for DRRA $(8, 4)$, -0.35 for CSA $(4, 4)$ against lag-1 autocorrelation; $p \approx 0.3$) — smoother series favor compression more; and (iv) a route-stability probe (one $(8, 4)$ run, seed 42, on a fixed validation batch) shows what the gradient-free selector does over training: the selected sets are *not* fixed (mean Jaccard overlap between the first and final epoch 0.83, with 37% of positions unchanged), while the routed *amount* is — the selected-set size holds at about 10 of 32 blocks and the route-score entropy stays flat (3.13) across all ten epochs — so membership follows the drift of upstream features and the routing amount does not. These probes are descriptive (n=1 seed for (i)–(ii) and (iv), n=12 for (iii)) and do not establish the mechanism.

Table 3: Compressed vs. full attention (25 paired seeds, frozen protocol; validation and test MSE in standardized units, lower is better). Δval and Δtest are relative changes against the Normal (full-attention) row, in percent, computed on the 4-decimal means (negative is better); "Seeds beating Normal" counts paired seeds where the cell's value is lower. Bold: best (lowest) mean in each column. Parameter counts are not equalized across the rows: the as-run Normal is a different attention class and carries $\sim$2% more full-model parameters, while the projection-matched reference Normal-r64 replaces its rank-128 output projection with the compressed cells' rank-64 bottleneck and carries $\sim$10% fewer.

Config	Val mean $\pm$ std	Test mean	Δval vs. Normal	Δtest vs. Normal	Seeds beating Normal
Normal	0.8389 ± 0.0814	1.0804	—	—	—
Normal-r64	0.7910 ± 0.0649	1.0281	-5.72%	-4.84%	15 / 25
$(4, 4)$	$\mathbf{0.6148} \pm 0.0704$	0.8771	-26.72%	-18.82%	24 / 25
$(8, 4)$	0.6190 ± 0.0827	**0.8440**	-26.22%	-21.89%	24 / 25

Table 4: Cross-dataset validation MSE (5 paired seeds, frozen protocol; pooled MSE in standardized units, lower is better). Models as rows (Normal = full attention; the compressed models are the (c_{KV}, c_R) cells); datasets as columns; bold: best mean per dataset column. The last row is the $(8, 4)$ cell's relative validation-MSE change against Normal (%; negative is better). Column abbreviations: Elec=Electricity, Traf=Traffic, Wea=Weather, Exc=Exchange, Sol=Solar-Energy, sSin=syn-sinusoids, sCP=syn-changepts, sDC=syn-delayedcopy.

Model	ETTh1	ETTh2	ETTm1	ETTm2	Elec	Traf	Wea	Exc	Sol	sSin	sCP	sDC
Normal	0.8416	1.9187	0.4285	1.5618	0.4047	0.5994	0.4460	**2.9800**	0.1899	**0.1227**	**5.9456**	**0.2888**
$(4, 4)$	**0.5909**	**1.7817**	0.4121	0.9326	**0.3775**	0.5907	0.4448	3.1364	**0.1693**	0.1276	5.9733	0.2894
$(8, 4)$	0.5967	1.8080	**0.3832**	**0.8053**	0.3808	**0.5874**	**0.4424**	3.2350	0.1722	0.1264	6.0013	0.2894
Δval $(8, 4)$ vs Normal	-29.1%	-5.8%	-10.6%	-48.4%	-5.9%	-2.0%	-0.8%	$+8.6\%$	-9.3%	$+3.1\%$	$+0.9\%$	$+0.2\%$

4.4 RQ3: cross-dataset coverage

To test whether the ETTh1 result is a single-dataset artifact, we repeated the three-way comparison (Normal vs. CSA $(4, 4)$ vs. DRRA $(8, 4)$) on eleven further datasets—three more ETT members, three large multivariate sets, two different-dynamics sets, and three mechanistic synthetic families—under the identical frozen protocol with the same five paired seeds (180 runs in total; per-seed tables in the released results manifests). Table 4 and Figure 4 summarize validation MSE by dataset. The ETTh1 column reflects the suite's five-seed run; the factorial (Table 2) reports the same configuration at 25 seeds, and the five pre-registered ETTh1 seeds are shared bit-identically between the two manifests.

Reading the table. Table 4 is transposed relative to the earlier ones: each *row* is a configuration (Normal, CSA $(4, 4)$, DRRA $(8, 4)$, plus a Δval row for DRRA $(8, 4)$ vs. Normal) and each *column* is one of the twelve datasets; a cell holds the mean validation MSE over the five paired seeds, with the best mean per dataset column in bold. Reading down a column compares the three models on one dataset—e.g., ETTm2 is the suite's largest win for DRRA $(8, 4)$ (-48.4%), while in the syn-delayedcopy column all three models agree within 0.001 and Normal's mean is best. Reading across a row compares one model across datasets, but *columns are not comparable to each other*: every dataset is standardized with its own train-only statistics, so MSE values only have meaning within a column. The final Δval row is scale-free (a ratio per dataset) and is therefore the row to compare across columns; it also carries the notable exceptions—positive entries on Exchange ($+8.6\%$) and on all three synthetic families ($+0.2\%$ to $+3.1\%$).

The compressed configurations beat full attention on mean validation MSE in 8/12 datasets each (DRRA $(8, 4)$ and CSA $(4, 4)$), but the advantage is *signal-dependent* rather than universal. Best-mean columns split evenly: 4 datasets each for CSA, DRRA, and full attention. The largest wins are on the ETT family (up to -48.4% on ETTm2) and on solar-energy; both compressed variants *lose* on all three synthetic families ($+0.2\%$ to $+4.0\%$) and on exchange-rate validation, and their test error is worse on weather (CSA $+11.2\%$, DRRA $+14.2\%$). This pattern is consistent with the compression-as-inductive-bias reading: pooling adjacent timesteps is nearly lossless on smooth, temporally redundant real-world series, while tasks requiring exact in-window content recall (the delayed-copy probe, known-period sinusoids) leave full attention with a slight edge. These observations remain descriptive: five paired seeds per dataset, no significance test, and the three rows are not parameter-equalized ($\sim$2% full-model spread; the same stance as Section 4.3).

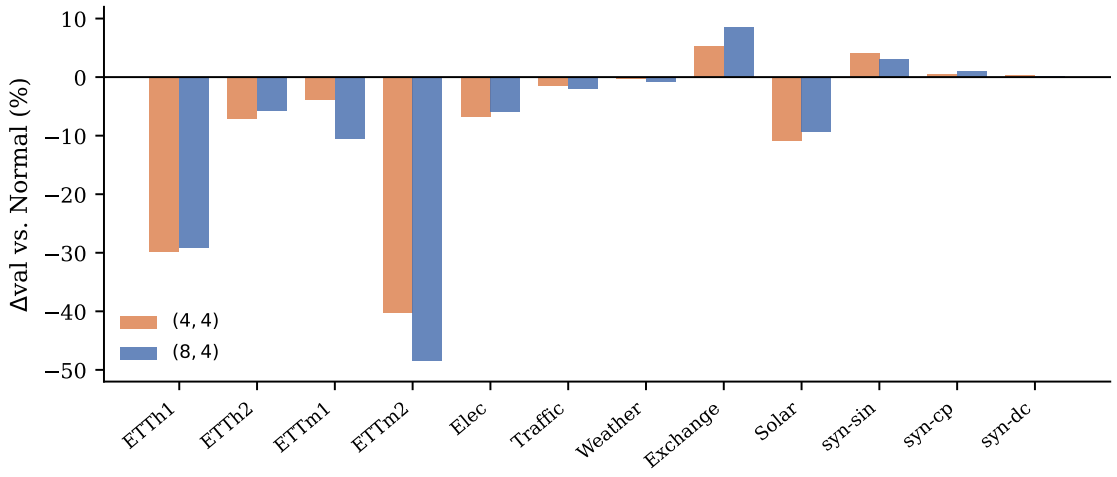


Figure 4: Validation-MSE change against full attention, by dataset (mean over 5 paired seeds): $(4, 4)$ (orange) and $(8, 4)$ (blue), the two compressed (c_{KV}, c_R) cells; Δval = relative change against the Normal (full-attention) baseline at the same seed, in percent, so negative (below the zero line) is better. Compression helps on the ETT family, solar, and electricity but slightly hurts on the synthetic probes and exchange-rate validation. Bars are means only (no error bars; per-seed tables in the released manifests). Dataset abbreviations: Elec=Electricity, syn-sin=synthetic sinusoids, syn-cp=synthetic change-points, syn-dc=synthetic delayed-copy.

4.5 External reference points

The comparisons above are internal to the family: every row shares one backbone and differs only in the attention module. To make the absolute MSE levels interpretable, we additionally ran two widely used published forecasting architectures under the *identical frozen protocol* (task, split, train-only normalization, optimizer, schedule, five paired seeds) as *reference points*, not as peer methods: **DLinear** (Zeng et al., 2023) (channel-shared configuration) and **PatchTST** (Nie et al., 2023) (official supervised ETTh1 configuration: patch 16, stride 8, $d_{\text{model}} = 16$, 3 layers, dropout 0.3, RevIN). Their edge over our minimal single-block backbone comes from their full architecture (patching, channel independence, decomposition; in DLinear's case no attention at all), not from their attention; the comparison is therefore confounded by design, and *no competitiveness claim is made or implied*. The purpose is calibration: what an interested reader needs in order to locate the family's absolute numbers. Because the protocol is held fixed while the architectures are not peers, the anchors are protocol-comparable but not method-comparable.

Table 5 reports the result. The anchors are decisively ahead on the real-world sets—most starkly, DLinear (49K parameters, no attention) outperforms every configuration of the family on ETTh1 validation and test. This is the LTSF-linear result of Zeng et al. (2023) reproduced under our frozen protocol, and it re-anchors the paper's reading: at this scale, the attention module is not the accuracy bottleneck, and the design-space structure measured in RQ1–RQ3 is a property of the attention family itself, orthogonal to backbone competitiveness. Consistent with this, the internal ordering of the family (compressed $>$ full attention on most datasets; decoupled-rate effects) is a statement within the shared backbone and is unchanged by the anchors' presence. PatchTST is reported only where its channel-independent training fits the single-machine CPU budget (the ETT family, exchange, and the synthetic probes); on the wide datasets (solar 137, electricity 321, traffic 862 channels) it is a *declared omission* ($\sim$3–13 h per seed on CPU), not a missing result (solar, electricity, traffic and weather omitted for the same budget reason).

The probes also bound the anchors' advantage from the other side: on two of the three mechanistic synthetic families—delayed-copy and known-period sinusoids—the ordering *inverts*, and both anchors fall behind even full attention (delayed-copy: Normal 0.289 vs. DLinear 0.408 and PatchTST 0.342; known-period sinusoids: Normal 0.123 vs. DLinear 0.251). The change-point family is excluded from this reading: there the anchors are far better than every family member (Normal 5.9456 vs. DLinear 0.0622, PatchTST 0.0489), so no inversion holds there. A PatchTST-backbone transfer with the same four cells grafted on finds all four tied within noise on ETTh1/ETTh2 (Appendix; forward latency DRRA 1.11x and (8,8) 1.27x vs CSA at tied accuracy); that transfer tokenizes the series into 31 patches, where the coarse-router cells can select every candidate index token, so the tie bounds the transfer

Table 5: External reference points under the frozen protocol (5 paired seeds; validation MSE in standardized units, lower is better). Reference points, not peer methods: the anchors carry their full published architectures (DLinear has no attention) while Normal and the two compressed cells share one minimal backbone (rows from Table 4). PatchTST is run where it fits the CPU budget and is a declared omission on the wide datasets (solar/electricity/traffic/weather; — = not run). Bold marks the better of the two external reference points (DLinear vs. PatchTST) in each row; the family columns (Normal and the compressed cells) are never bolded, so where a family cell has the lower mean the bold anchor is not the row minimum.

Dataset	DLinear	PatchTST	Normal	$(4, 4)$	$(8, 4)$
ETTh1	**0.3521**	0.3603	0.8416	0.5909	0.5967
ETTh2	0.2753	**0.2653**	1.9187	1.7817	1.8080
ETTm1	0.2568	**0.2517**	0.4285	0.4121	0.3832
ETTm2	0.1594	**0.1553**	1.5618	0.9326	0.8053
Electricity	**0.1514**	—	0.4047	0.3775	0.3808
Traffic	**0.3499**	—	0.5994	0.5907	0.5874
Weather	**0.4575**	—	0.4460	0.4448	0.4424
Exchange	0.3000	**0.2063**	2.9800	3.1364	3.2350
Solar	**0.1617**	—	0.1899	0.1693	0.1722
sSin	0.2509	**0.1969**	0.1227	0.1276	0.1264
sCP	0.0622	**0.0489**	5.9456	5.9733	6.0013
sDC	0.4078	**0.3419**	0.2888	0.2894	0.2894

claim from below rather than establishing that the rate effects vanish on stronger backbones (Appendix C). Tasks requiring exact in-window content recall are exactly where attention helps relative to linear or patched architectures—consistent with the compression-as-inductive-bias reading of Section 4.4, and worth further study.

4.6 Statistical treatment

Statistical status is reported separately for the two levels of the study.

Factorial (ETTh1, n=25 paired seeds; full analysis log released with the source). At 25 paired seeds, paired Wilcoxon signed-rank tests on validation MSE are powered: *every* compressed cell beats full attention, and the result survives Holm correction across the five pre-registered contrasts (Holm-adjusted $p \leq 3.3 \times 10^{-6}$ for all four cell-vs-Normal contrasts). The projection-matched comparisons of Section 4.6 are robustness analyses of the same claim, not additional confirmatory tests: they are reported with paired 95% intervals and unadjusted Wilcoxon p-values, and the pre-registered family remains the five contrasts of the pre-registration (the four cell-vs-Normal comparisons plus DRRA-vs-CSA). The pre-registered H1 contrast (per-seed routing effect minus KV effect) is likewise significant (exact Wilcoxon $p = 0.0074$; paired t $p = 0.0107$). The DRRA-vs.-equal-rate edge is *not* significant on validation (DRRA vs. CSA: exact Wilcoxon $p = 0.62$; 16/25 head-to-head; no cell best of all four in more than 8 of 25), and "best on mean" is no longer claimed at all. On *test* MSE the same contrast favors DRRA (mean -0.0331, 95% CI $[-0.070, +0.004]$, exact Wilcoxon $p = 0.032$; 17/25 head-to-head; best of four in 12/25 for DRRA)—a secondary, unadjusted contrast reported alongside the pre-registered validation analysis. The n=25 set is a post hoc merge of the five pre-registered

seeds with a 20-seed power extension run under the byte-identical protocol (plus the corrected-mapping re-runs of three cells under the same protocol); the seed list was fixed in the extension's design before any extension run, and the merge was planned at that time.

Cross-dataset suite (n=5 paired seeds per dataset; analysis log released with the source). Per-dataset, the three comparisons (DRRA vs. Normal, CSA $(4, 4)$ vs. Normal, DRRA vs. CSA) were tested with exact paired Wilcoxon signed-rank tests; *nothing survives Holm correction across the 36 tests*, and with n=5 the two-sided floor is $p = 0.0625$, so no per-dataset effect can reach two-sided $p < 0.05$ (the closest raw values are etth1 and solar at 0.0625 in favor of DRRA $(8, 4)$). The two exact-recall probes reach significance against compression one-sided (exact $p = 0.03125$: all five seeds favor full attention on syn-sinusoids and syn-delayedcopy); syn-changepoints is not significant in either direction. Across datasets, paired sign tests on the 12 dataset means are not significant: 8/12 ($p = 0.39$) for DRRA $(8, 4)$ and 8/12 ($p = 0.39$) for CSA $(4, 4)$. The strongest scale-free aggregate statement is the bootstrap 95% CI on the mean of the per-dataset Δval vs. Normal, which excludes zero for both variants (CSA $(4, 4)$: -7.5% $[-15.8, -0.9]$; DRRA $(8, 4)$: -8.3% $[-17.5, -0.8]$). The cross-dataset pattern is therefore directionally consistent but not statistically established per dataset: the ETTh1 factorial, not the suite, is where significance is established (25 paired seeds versus 5 per dataset). Deferred items, with written justification in the project plan: the mixed-effects model (Section 4.2), HAC/block-bootstrap corrections (the analyses are at the seed level, where the paired runs are independent, so stride-1 window overlap does not invalidate the paired tests), and TOST/ROPE (the paper makes no equivalence claims).

Sequence-length scaling and complexity. Figure 5 sweeps the lookback $L \in \{128, \ldots, 4096\}$ at batch 1 and 32. At batch 32 DRRA $(8, 4)$'s advantage over Normal grows from -32% at L=512 to -56% at L=2048 (869 vs. 1984 ms/batch, a 2.3× speedup), and at batch 1/L=4096 it reaches -63% (86 vs. 235 ms)—while at $L \leq 256$ and batch 1 Normal is fastest (fixed overheads again). The empirical log-log slopes ($L \geq 512$) are 1.40 (DRRA $(8, 4)$), 1.42 (CSA $(4, 4)$), and 1.72 (Normal) at batch 32: consistent with the complexity analysis of Section 3.4—the routing term $O(L^2/c_R)$ keeps every cell asymptotically quadratic, with a reduced constant for the compressed cells, and Normal approaches the quadratic slope fastest. We therefore do *not* claim subquadratic scaling. At the extreme, batch-32/L=4096 full attention requires a $\sim$8 GB score tensor and was skipped on the 16 GB machine. That $\sim$8 GB wall is a property of our materializing implementation, not of dense attention as such: fused kernels (e.g. FlashAttention (Dao et al., 2022)) avoid materializing the score tensor and move the wall elsewhere—so the memory advantage reported here is implementation-relative, and GPU transfer may erase it. All models process the frozen task in under 2 ms per sequence at batch 32 (mean 1.6–1.8 ms/seq at L=256; the $L \geq 512$ region of the scaling panel is where attention, not fixed overhead, dominates per-batch time).

Compute and memory accounting. Analytic forward MACs per sequence (L=256, dominant terms; closed-form accounting in the released benchmark scripts): Normal 76.9M, CSA $(4, 4)$ 63.3M, DRRA $(8, 4)$ 61.2M — DRRA $(8, 4)$ needs 20% fewer MACs than Normal. Three caveats bound that number. First, the as-run rows are *not* projection-matched: the dense baseline carries a rank-128 output projection (21.0M) where the compressed cells use the equalized rank-64 bottleneck (10.5M), and that one asymmetry is 10.5M of the 15.7M whole-model gap. Matching the projection ranks leaves a 7.8% whole-model difference, while the attention path itself falls 30.9% and the attention core falls 80.9% (Section 4.8). We therefore retrained the dense baseline with the matched projection (25 paired seeds, same protocol): matching the rank makes the dense model *better*, not worse (0.7910 vs 0.8389 validation MSE; paired mean -0.048, 95% interval $[-0.088, -0.008]$), so the as-run Normal was not a conservative reference. Every compressed cell still beats the matched baseline, with paired validation deltas -0.110 to -0.178 MSE whose 95% intervals all exclude zero (21–25 of 25 seeds per cell), and the ordering is the same on test MSE. The accuracy advantage is therefore not a projection artifact: matching the projection removes the MAC wedge and leaves the accuracy gap. Second, the wrapper MLP (33.9M) is the largest single block for *every* model, so attention is not the dominant compute at this scale. Third, measured latency differences (-6%) are smaller than the MAC difference because dense-masked CPU GEMMs are efficient relative to the extra compressor/indexer work. And MACs are not energy: arithmetic counts omit memory traffic and hardware-specific efficiency, so any of these MAC gaps is a work statement, not an energy claim (the deployment reading in Section 6 is explicitly conditional on hardware where attention dominates). Peak resident memory during a batch-32 forward burst grows by $\sim$0.3 GB (process watermark; parameter memory is only $\sim$1.4 MB per model), and at batch-32/L=4096 the full-attention score tensor ($\sim$8 GB) exceeds the 16 GB machine. Training time over the fixed 10-epoch ETTh1 budget (parsed from the sweep log; medians of 5 runs): Normal 474 s, CSA $(4, 4)$ 511 s, DRRA $(8, 4)$ 459 s — one DRRA run (1417 s) overlapped a machine-sleep window and is excluded from the median. All latency/compute measurements are single-machine (Apple M2, fp32, 8 threads); cross-hardware transfer (Linux x86, GPU) and model depth/width sweeps remain future work (Appendix future-work item 4).

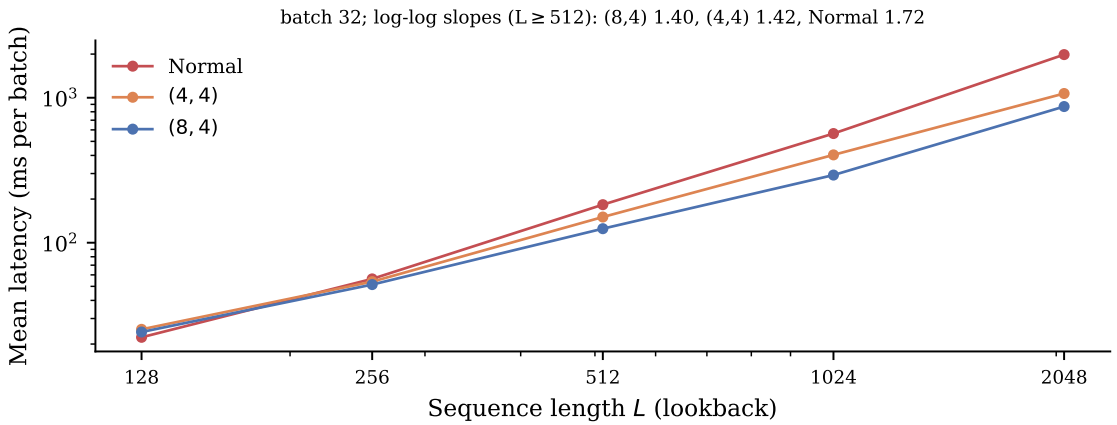


Figure 5: Forward-pass latency vs. lookback length L (log-log; batch 32 = 32 input windows per forward pass; mean over per-round medians); curves: Normal (full attention), $(4, 4)$, $(8, 4)$; slope = fitted exponent on the log-log axes (2 = quadratic). All cells are asymptotically quadratic (routing term $O(L^2/c_R)$, with c_R = routing resolution in positions per index token); the compressed cells have smaller constants and lower empirical slopes (1.40/1.42 vs. 1.72 for Normal at batch 32), and batch-32/L=4096 full attention exceeds practical memory (see text). All measurement and regeneration logs are released with the source (batch-1 sweep in the same logs).

4.7 Efficiency

Table 1 gives the structural efficiency story: DRRA $(8, 4)$ attends to 48 columns per query (5.22× fewer complete columns than Normal) at 95,652 attention parameters—*fewer* than Normal's 102,496—and a full-model count 2.0% below Normal's. Routing cost is identical for the two c_R=4 cells regardless of c_{KV}, which is the deliberate trade-off: pay the full fine-routing cost, halve the attention cost.

Table 6 reports wall-clock forward-pass latency measured *now* under the frozen protocol on the development machine, using the alternating head-to-head protocol (all three models timed in rotation within each round, so each round shares the same thermal/contention window; 25 rounds of 30 passes after warm-up; mean over per-round medians with 95% percentile bootstrap CIs, plus p90/p99 tails). DRRA $(8, 4)$ has the lowest mean latency (49.3 ms vs. 52.5 ms for Normal, -6.1%; CSA $(4, 4)$ -0.7%), and the ordering $(8, 4) < (4, 4) <$ Normal replicated across a second run, but the 95% CIs overlap and the level shifts by $\sim$6–9% between runs (thermal and contention effects), so we report the latency comparison as *directional* only. Paired within-round analysis tightens this without changing the stance: in a 25-round round-robin rerun under the frozen protocol (released audit log), DRRA $(8, 4)$ beats Normal by -9.0 ± 1.7 ms per paired round and CSA $(4, 4)$ by -5.2 ± 0.9 ms (95% CIs exclude zero *in that run*); absolute levels still drift across runs, so directional-only stands. A last-query audit reframes the comparison further: the forecast uses only the last position (Section 3.1), and recomputing latency for that forecast-required path (bit-exact to $\sim 5 \times 10^{-7}$) gives Normal 2.6–3.9 ms, CSA $(4, 4)$ 6.2–9.5 ms, DRRA $(8, 4)$ 5.8–8.6 ms per batch-32 pass—ranges span two runs showing the same level drift. Under last-query accounting the dense baseline's last-query path is the fastest, because it pays no compressor/indexer work: Table 6 compares full-layer costs, both costs are reported, and efficiency claims are stated against the last-query baseline. The exploratory cells were timed the same way: full-layer $(16, 4)$ and $(32, 4)$ run at $1.22\times$ Normal (37.3–37.4 ms vs. 45.6 ms), barely ahead of $(8, 4)$ at $1.18\times$ — wall-clock plateaus past $(16, 4)$ because compressor work and dense masking dominate the shrinking column count, so further compression buys accuracy-neutrality, not speed. Notably, the structural 5.22× column reduction does not translate into proportional wall-clock gains at this scale: dense-masked CPU GEMMs are efficient and the compressor/indexer add overhead. These are single-machine CPU measurements (fp32, 8 threads); cross-hardware transfer (Linux x86, GPU) remains follow-up work (Appendix future-work item 4). The pilot-era speed figures from earlier stages of this project remain superseded and are not used here.

Batch-size scaling. Figure 7 shows how the comparison changes with the number of input windows processed per forward pass (mean ms per sequence). At batch size 1 the fixed per-model overheads (compressor, indexer) dominate and Normal is fastest (2.5 vs. 3.3 ms/seq for DRRA $(8, 4)$, $+30.8\%$). From batch 8 onward the compressed cells run

Table 6: Forward-pass latency under the frozen protocol on the development machine (Apple M2, 8 cores, 16 GB; CPU PyTorch 2.6.0, fp32 = single precision, 8 threads). Latency is the wall-clock time of *one forward pass over a batch of 32 input windows* (L=256 lookback, H=96 forecast horizon, F=7 input channels); per-sequence latency at this batch size is mean/32 ($\approx$1.5 ms/seq for the $(8, 4)$ cell); Δmean vs. Normal is the relative mean-latency change against the full-attention baseline (negative = faster). Alternating head-to-head protocol (25 rounds $\times$ 30 passes, warm-up excluded); mean over per-round medians (95% percentile bootstrap CI = confidence interval, in brackets), with p90/p99 tails (90th/99th-percentile latency); throughput from the mean pass time. Measurement log released with the source.

Model	Mean (ms)	p90 (ms)	p99 (ms)	Δmean vs. Normal	Throughput (batches/s)
Normal	52.5 [50.4, 54.7]	59.4	65.0	—	19.1
$(4, 4)$	52.1 [49.2, 55.4]	63.5	71.5	-0.7%	19.2
$(8, 4)$	49.3 [47.2, 51.7]	56.5	65.9	-6.1%	20.3

at or below Normal, and at large batches the attention-column reduction takes over: the DRRA $(8, 4)$ mean advantage over Normal is -6.1% at batch 64, -5.1% at 100, -10.5% at 128, and -21.1% at 256 (3.49 vs. 4.42 ms/seq). At the mid-range batches 16–32 the two are within run-to-run noise (the scaling run shows -0.1 to $+0.2\%$ there while the canonical Table 6 run shows -6.1% at batch 32). The overall pattern is the expected signature of amortized overheads: the compression machinery costs fixed work per pass, while the saved attention columns scale with the batch.

Synthesis: is the dual-rate design better? Each table speaks to one axis of "better." Table 1 establishes the *structural* side: DRRA $(8, 4)$ needs 48 attention columns (5.22$\times$ fewer complete columns than Normal), slightly fewer parameters than Normal, and the same fine-routing candidate count as the equal-rate cells. Tables 2–4 establish the *accuracy* side: DRRA $(8, 4)$ beats Normal on mean validation MSE in 8/12 datasets, and the top three factorial cells on ETTh1 sit within seed noise of each other (no single best cell), with effects that are directional and, on the synthetic probes and some datasets, reversed. Table 6 and Figure 7 establish the *measured-latency* side: DRRA $(8, 4)$ has the lowest mean latency at batch sizes $\geq$ 64 (to -21% at 256) but is not statistically separated from Normal at batch 32, and is slower at batch 1. Read together, the evidence favors DRRA $(8, 4)$ on every axis on both backbones (Appendix transfer-speed figure shows the same pattern on PatchTST)—geometry, parameters, accuracy on most datasets, large-batch latency—but none of the advantages clears a significance or noise threshold on its own, and the accuracy advantage is dataset-dependent. We therefore state the conclusion as *directional superiority under this protocol on most datasets*, not a universal or statistically established claim.

Combined performance–speed view. Figure 6 shows performance and speed *together*, without collapsing them into a single score: the y-axis is the mean test *loss on ETTh1* (the frozen-task benchmark dataset, where all three models share the same standardized scale; per-dataset losses across the suite are *not* averaged, because each dataset is standardized with its own train-only statistics), and the x-axis is *latency* at the frozen batch 32 (lower latency = faster = further left). The left panel uses the *forecast-required* path—the deployment-relevant accounting, in which only the last position is evaluated (Section 3.1)—and the right panel reports the full-layer path separately (Table 6 carries the full-layer numbers, including batch 256). Both axes are zoomed to the measured range so differences are visible; points are five-seed means, with no error bars shown. The two accountings rank speed differently, and the reversal is the point: on the forecast-required path Normal is fastest (2.6 vs. 6.2 and 5.8 ms for CSA $(4, 4)$ and DRRA $(8, 4)$), because it pays no compressor/indexer work, while on the full-layer path DRRA $(8, 4)$ is fastest (31.7 vs. 37.0/40.7 ms; means of one audited 25-round round-robin run). Accuracy is the same in both panels: **DRRA** $(8, 4)$ has the lowest mean test loss on ETTh1 (0.799 vs. 0.883 for CSA $(4, 4)$ and 1.081 for Normal). The honest reading is a trade-off, not dominance: under forecast-required accounting the dense baseline is the fastest per forecast and the compressed cells are more accurate on ETTh1, whereas under full-layer accounting DRRA $(8, 4)$ is lower-left (best on both axes). The full-layer view is retained because it is the honest comparison for architectures whose head needs all temporal outputs; the efficiency claims in this paper are stated against the forecast-required baseline. Both panels remain descriptive (five seeds per dataset, no significance tests, single machine).

4.8 Scaling forecast and energy projection

The frozen task fixes the context at L=256, but deployed attention systems increasingly run at context lengths of thousands to hundreds of thousands of tokens. Re-measuring the protocol at every length of interest would be costly and, more importantly, unnecessary: the cost of this family is governed by a column geometry that is known in closed

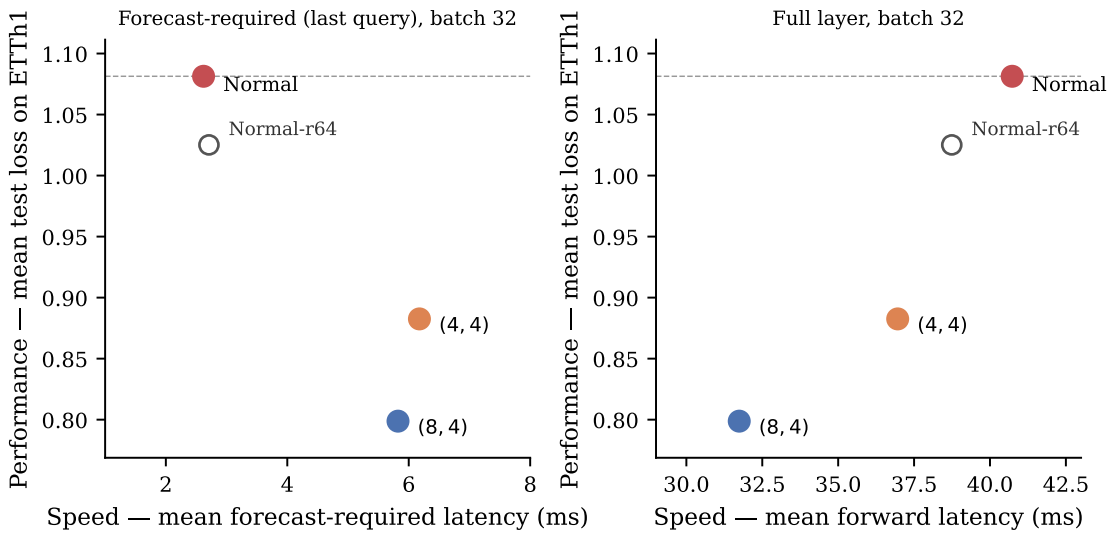


Figure 6: Performance vs. speed: y-axis = mean test loss (MSE, mean squared error) on ETTh1 (mean over the five paired seeds; lower is better); x-axis = mean latency in milliseconds (lower = faster = further left), batch 32; left: the *forecast-required* path (last position only; the deployment-relevant accounting), right: the full-layer path (retained for architectures whose head needs all temporal outputs). On the forecast-required path Normal is fastest; on the full-layer path $(8, 4)$ is fastest; accuracy is identical in the two panels. The open marker is the projection-matched dense baseline (Section 4.6); its latency is anchored to the released Normal measurement by a same-run paired timing, because machine load shifts all models together. Measurement logs released with the source.

form and has already been audited at L=256 (Section 3.3). We therefore use the audited component accounting to *forecast* attention work across the context range $L \in [256, 2^{17}]$ and to translate that forecast into an explicit energy projection (Figure 8). No model is retrained and no new run is performed; the forecast is a statement about the geometry this paper has already measured, and Appendix D gives its full derivation, sensitivity analysis, and audit.

Three savings measures. We keep three quantities separate, because they answer different questions and differ by more than an order of magnitude at some scales. The *attention matrix* is the number of score columns each query reads: L for dense attention, and $L/c_{KV} + W + 1$ for a compressed cell. The *attention path* adds the two costs the family introduces—the router, which compares each query against L/c_R index tokens, and the two-series compressors—so that a compressed cell is charged for everything it executes. The *whole layer* adds the non-attention block (projections and MLP), which the cell definition leaves unchanged. One column read costs $2d_h$ multiply–accumulate operations (MACs), one score and one value read, and every count is fixed by the design and evaluated analytically; Table 7 collects the notation with the frozen values.

For a sequence of length L the attention core of the dense baseline performs $W_N(L) = 2hd_hL^2$ multiply–accumulate operations (scores and values; the study's frozen geometry is h=4, d_h=32, d=128, W=16, with $d = hd_h$), while a compressed cell with rates (c_{KV}, c_R) performs

$$W_C(L; c_{KV}, c_R) = 2hd_hL\left(\frac{L}{c_{KV}} + W + 1\right) + d\frac{L^2}{c_R} + 2Ld\,(64 + d_h), \tag{2}$$

where the three terms price, in order, the score–value computation over the fixed column set, the router, and the two-series compressors. The first covers the L/c_{KV} compressed blocks, the W uncompressed window positions, and the sink column per query, at $2d_h$ MACs per column per head, where the factor L counts queries, each paying for the same menu; the second scores all L/c_R index tokens per query with every head of the shared routing query, at $hd_h = d$ MACs per comparison (the routing query is the main multi-head query, so one head's dot product per token pair would under-count by h; the executed tensor shapes are recorded in the released audit log), leaving the routing cost quadratic in L with the reduced constant $1/c_R$; the third is the per-token cost of building both compressed series, $2d(64 + d_h)$ MACs, the only term linear in L. That linear term dominates at short context—6.29M of the 11.60M attention-path

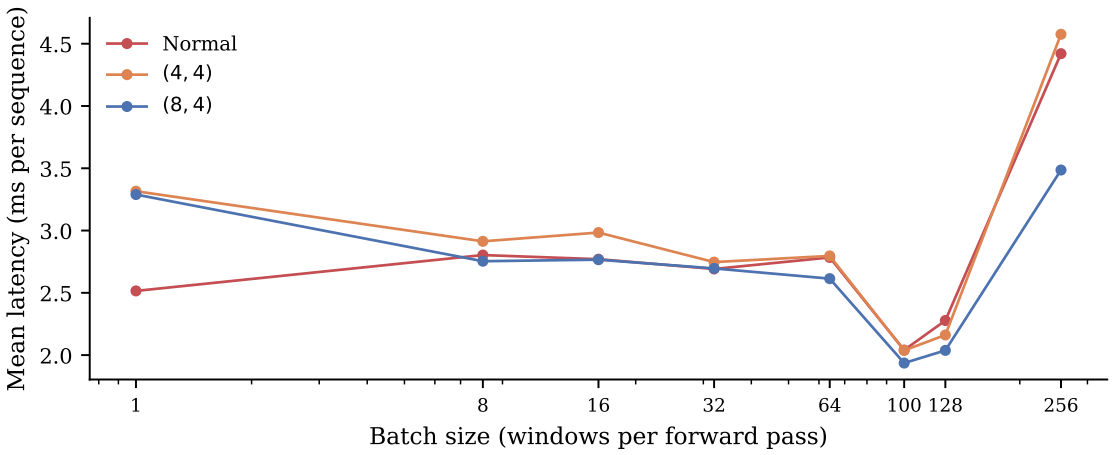


Figure 7: Mean forward-pass latency per sequence vs. batch size (number of input windows per forward pass) under the frozen protocol (development machine, fp32 = single precision, 8 threads; mean over per-round medians, 15 rounds $\times$ 15 passes per model per batch size). At batch size 1 the fixed overheads of the compressor/indexer dominate and Normal is fastest; from batch 64 onward $(8, 4)$ is clearly fastest, with the gap widening to -21% at batch 256. Measurement log released with the source.

Table 7: Notation for the projection, with values for the frozen configuration. The rates c_{KV} and c_R take the values listed per cell in Table 1.

Symbol	Meaning	Value
L	context length: past positions each query may read	256
c_{KV}	positions per compressed key–value block	4 or 8
c_R	positions per routing index token	4 or 8
W	uncompressed sliding window	16
$+1$	sink column	1
h	attention heads	4
d_h	head dimension	32
d	hidden size, equal to $h\, d_h$	128
64	compressor projection width	64

MACs of DRRA $(8, 4)$ at L=256 (the router contributes 2.10M), hence its 30.9% saving there—and is asymptotically negligible. The corresponding attention-path saving,

$$S(L) = 1 - \frac{W_C(L)}{W_N(L)}, \qquad S_\infty = \lim_{L\to\infty} S(L) = 1 - \left(\frac{1}{c_{KV}} + \frac{1}{2c_R}\right), \tag{3}$$

is the fraction of attention-path work removed at length L. Here $W_N(L) = 2hd_hL^2$ is the dense attention work introduced above, and $W_C(L)$ abbreviates the compressed attention-path work $W_C(L; c_{KV}, c_R)$ of Equation 2, so the ratio inside $S(L)$ is the share of dense work the compressed path still spends. In the limit only two terms survive—the retained columns perform $1/c_{KV}$ of the dense attention work and the router contributes $1/(2c_R)$—because these are the coefficients of the two L^2 terms of Equation 2, while the window offset and the compressor cost are $O(1)$ and $O(L)$ against the $O(L^2)$ attention terms; for DRRA $(8, 4)$ this leaves $1/8 + 1/8 = 25.0\%$, or 75.0% removed. Dividing Equation 2 by W_N exposes the same decomposition at finite context: with the frozen geometry ($d = hd_h$) the compressed path pays four fixed shares of dense work, $1/c_{KV} + (W+1)/L + 1/(2c_R) + (64+d_h)/L$, of which the two $1/L$ shares shrink as context grows; in the evaluated form the compressed path reads $W_C(L) = 256\, L\, (L/c_{KV} + 17) + 128\, L^2/c_R + 24{,}576\, L$, with $256 = 2hd_h$, $17 = W + 1$, $128 = hd_h = d$, and $24{,}576 = 2d(64 + d_h)$. At the frozen context the savings are modest—18.4% to 40.2% across the four cells, because the linear compressor term dominates at L=256—but they grow with context and saturate at the limit: at L=32,768 the four cells remove 62.2%, 74.7%,

Table 8: Structural attention-path savings (percent of dense-attention work removed) as a function of context length L, from Equation 3; columns are the (c_{KV}, c_R) rates (positions per block/token). The frozen-task row anchors the measured operating point; the limit row is the closed-form asymptote S_∞. All values are analytic projections of the audited component accounting, not measurements; the accompanying script reproduces the released per-component totals exactly before drawing them.

Context L	$(4, 4)$	$(8, 4)$	$(16, 4)$	$(32, 4)$
256 (frozen task)	18.4%	30.9%	37.1%	40.2%
4,096	59.7%	72.2%	78.5%	81.6%
8,192	61.1%	73.6%	79.9%	83.0%
32,768	62.2%	74.7%	80.9%	84.0%
131,072	62.4%	74.9%	81.2%	84.3%
$L \to \infty$ (S_∞)	62.5%	75.0%	81.2%	84.4%

80.9%, and 84.0% of attention-path work, respectively, within a few tenths of a percentage point of S_∞, approaching it from below with the gap set by the compressor term (Table 8). The map over the design plane (Figure 8d) makes the asymmetry quantitative: moving c_{KV} from 4 to 8 at c_R=4 adds 12.5 percentage points of asymptotic saving, while moving c_R from 4 to 8 at c_{KV}=8 adds 6.2—so the accuracy-favored axis is also the larger saving lever, and keeping the router fine costs the smaller share (Figure 8b, dashed lines).

Whole-layer projection. A saving on the attention path is not the same as a saving on the layer, because the non-attention block is unchanged by the cell definition. Holding that block fixed at the standard per-token cost of $12d^2$ multiply–accumulate operations ($4d^2$ for the query, key, value, and output projections and $8d^2$ for the MLP; at d=128 this reproduces the audited compressed-cell non-attention accounting to within 1.5%, 49.60M against 50.33M operations per sequence), the projected whole-layer saving is

$$S_{\text{layer}}(L; d) = \frac{2dL - \left[2d\left(\frac{L}{c_{KV}} + W + 1\right) + d\frac{L}{c_R} + 2d(64 + d_h)\right]}{2dL + 12d^2}. \tag{4}$$

Both terms are per-token: the numerator is the attention-path work removed, and the denominator the dense layer cost—attention $2dL$, which is $2hd_hL^2$ divided by L since $d = hd_h$, plus the unchanged $12d^2$ block—so the ratio is the fraction of whole-layer work removed. At a 32k-token context, DRRA $(8, 4)$ removes 68.3% of whole-layer work at d=512, 62.9% at d=1024, and 42.7% at d=4096 (an 8B-class width), while the $(32, 4)$ probe removes 76.8%, 70.8%, and 48.0%; the fraction shrinks as d grows because attention is then a smaller share of layer work. For the study's own backbone (d=128) the projected whole-layer saving is 50.5% at L=2048 and 60.8% at L=4096.

The deeper KV cuts continue the trend with sharply diminishing returns at the studied scale. The asymptotic attention-path saving rises from 75.0% for $(8, 4)$ to 81.2% and 84.4% for $(16, 4)$ and $(32, 4)$; at the frozen context the same cuts add 6.2 and 9.3 points (37.1% and 40.2% against 30.9%), because the compressor cost is invariant to c_{KV}, and the measured full-layer speedup plateaus at 1.22× for both probes against 1.18× for $(8, 4)$—a difference within the run-to-run drift documented for these measurements. Accuracy gives no counterpart reason to promote the deeper rates: the probes are exploratory (post hoc, outside the pre-registered grid) and their spread is within seed noise; $(16, 4)$ has the best mean of the three on both endpoints (0.5810 validation and 0.8100 test against 0.6190/0.8440 for $(8, 4)$ and 0.6122/0.8586 for $(32, 4)$, 25 seeds each), yet $(32, 4)$—the deepest cut—improves on $(16, 4)$ on neither, so coarser KV compression is not accuracy-monotone and neither probe carries a confirmatory claim (Appendix C). We therefore keep $(8, 4)$ as the named dual-rate corner and treat the deeper rates as boundary probes. They remain relevant for extreme-context deployments, where the column budget of Figure 8a dominates; that evaluation is left to future work.

From work to energy. To connect these counts to energy without claiming an energy measurement, we adopt the first-order model $E(\theta) = \sum_k \varepsilon_k\, n_k(\theta)$, in which energy is proportional to arithmetic work $n_k(\theta)$ with per-operation coefficients ε_k. Here θ denotes the configuration (cell and context length), k indexes operation type, $n_k(\theta)$ is the number of operations of type k (the counts of Equation 2), and ε_k is the energy per operation of type k, a hardware rather than a model property. With a single coefficient, $E \propto W$, so the savings in Equations 3 and 4 *are* fractional energy savings: the coefficient is common to the baseline and to the compressed model and cancels in the ratio, leaving the headline projections independent of processor and process node under the stated model. Figure 9 shows the composition this leaves at the study operating point (L=2048): compression shrinks the attention core from 72.7%

of dense per-token work to 18.8% (CSA) and 9.7% (DRRA), the router and compressor add 12.5%, and the projections-and-MLP block is untouched at 27.3%, so the whole-model saving is smaller than the attention-path saving—41.4% versus 57.0% for CSA and 50.5% versus 69.5% for DRRA. Two consistency checks on the work-to-time linkage that any such model requires are available from the systems measurements. First, the measured forward latencies rank with the analytic attention-path work across the audited lookback sweep (Spearman $\rho = 1.000$ at batch 32 and 0.992 at batch 1). Second, at the operating points where the two are commensurate, the whole-layer projection tracks the measured end-to-end saving: at L=2048/batch 32 it predicts 50.5% against 56.2% measured, and at L=4096/batch 1 it predicts 60.8% against 63.3% measured. The measured saving now exceeds the arithmetic projection by 2.5–5.7 points, the opposite of what a pure work model predicts; the likely source is that the dense baseline is memory-bound at these contexts (the materialized score tensor dominates its time), so its wall-clock cost is larger than its MAC count—an effect the work model deliberately excludes, and the same implementation-relative memory advantage that Section 4.7 already reports. Two sensitivities bound the model's assumptions (Appendix D): reweighting the router and compressor terms by a factor of 0.5 leaves the 32k DRRA projection at 81.1% and by a factor of 2 at 61.9% (central case 74.7%), with the direction unchanged; and the executed router is the expensive arrangement—it scores every index token with every head of the shared query, and a single-head indexer (as in a Lightning-Indexer-style design, which this module does not use) would raise the 32k DRRA projection from 74.7% to 84.0% and the asymptote from 75.0% to 84.4%.

What is *not* claimed is equally explicit. These are model-based projections from arithmetic counts, not energy measurements (Section 4.7); they model arithmetic work only, so memory stalls, frequency scaling, and leakage are outside the model; and they inherit the single-machine, CPU-scale scope of the underlying audit. The value of the projection is that its leading term is a hardware-independent ratio—the work reduction—which any energy evaluation of this family must account for, and which the measurements above anchor in direction.

5 Discussion

5.1 The decoupling insight (directional)

The cleanest reading of the factorial is that the two axes play different roles on ETTh1, and that the roles are conditional: *coarse routing costs accuracy where the KV content is fine (the $(4, 8)$ cell), while coarser KV content is mildly beneficial to accuracy and buys efficiency*. The control logic is internal to the design: the DRRA $(8, 4)$ cell (coarse content, fine router) matches or beats the equal-rate cells with roughly three-fifths the attention columns (49 vs. 81 complete columns), while the $(4, 8)$ cell (fine content, coarse router) is the worst cell despite having the same columns as CSA $(4, 4)$ — and at c_{KV}=8 the finer router adds nothing on the mean ($(8, 8)$ 0.6128 vs. DRRA 0.6190). The per-seed consistency (20/25 for KV, 16/25 for routing), the pre-registered H1 contrast (exact Wilcoxon $p = 0.0074$), and the significant interaction ($p = 0.014$; the axes are not additive) establish this as a *statistically supported sign structure with non-additive axes on ETTh1*, whose interaction is carried by the experiment's worst cell rather than by a general benefit of fine routing; the effect sizes remain modest relative to seed noise (both effects $\sim$0.03 vs. within-cell std $\sim$0.070), no cell is best on validation (top three tied within noise; DRRA-vs.-CSA $p = 0.62$), and multi-dataset confirmation of the full factorial is still future work before this can be stated as a general design principle. An exploratory (16,4) probe (best mean 0.5810, ahead of Normal 25/25) and a further (32,4) probe (0.6122, inside the top group) suggest the KV-rate profile stays flat-to-beneficial out to 32:1 on ETTh1, with 4 KV blocks as the sensible floor.

5.2 Why might compression help on this task?

We do not have a verified explanation for the compressed $>$ full-attention outcome, and we explicitly do not claim one. Candidate (unverified) explanations include: compression acting as an inductive-bias regularizer; the block structure effectively smoothing the attention field over multi-step correlations; and the fixed 10-epoch budget interacting with optimization landscape differences. The cross-dataset pattern (Section 4.4) is broadly consistent with the regularization reading—compression helps most where adjacent timesteps are most redundant (ETT family, solar, electricity) and slightly hurts on tasks requiring exact in-window recall (the synthetic probes)—but it remains descriptive evidence, not a verified mechanism. The pre-registered hypothesis program (H1 is defined in Section 4.2) is designed to separate the candidates, and its current status is: **H1** (coarsening the router deteriorates accuracy more than coarsening KV compression) supported on ETTh1 (exact Wilcoxon $p = 0.0074$) but not replicated on the three cross-dataset factorial replications (all non-significant, signs mixed; the indexer receives no task gradient, so granularity effects are architectural); **H2** (the dual-rate corner matches the equal-rate cells on accuracy while using fewer attention columns), whose latency leg is supported directionally by the systems measurements and whose accuracy leg is a tie on validation

(no cell best; DRRA-vs.-CSA $p = 0.62$) with DRRA ahead on test (secondary contrast, $p = 0.032$) and 8/12 suite datasets beating Normal on mean; **H3** (dense masked attention wins in small-work regimes but loses in large-work regimes), whose crossover shape is observed on a single machine; **H4** (the compression advantage depends on signal properties such as smoothness and redundancy), consistent with the cross-dataset pattern; and **H5** (a distilled surrogate can replace attention at inference without an accuracy loss), which is out of scope for this paper—deferred to a separate publication (Appendix B). The mechanistic analyses that would separate the explanations (routing recall, lag localisation) remain future work.

5.3 Limitations

- **Dataset breadth, not universality.** The full 25-seed factorial is single-dataset (ETTh1); a five-seed replication on three contrasting datasets did not reproduce its sign structure (Section 4.2). The three-way comparison spans twelve datasets, where the compressed advantage is signal-dependent (Section 4.4): wins on the ETT family, solar, and electricity; small losses on the synthetic probes, exchange-rate validation, and weather test error. No cross-scale claim is made (CPU-scale models only), and horizons beyond $H{=}96$ with lookbacks beyond $L{=}256$ are not yet evaluated.
- **Descriptive statistics.** Five paired seeds; per-dataset paired tests cannot reach significance at this sample size (two-sided floor $p = 0.0625$) and nothing survives Holm correction (Section 4.6); "best on mean" and "directional" are used advisedly. Figures show means only; per-seed tables are in the released manifests. No equivalence tests are run because no equivalence claims are made.
- **External architectures as reference points only.** DLinear and PatchTST were run under the frozen protocol to calibrate absolute error levels (Section 4.5); they are not peer methods (different architectures, and DLinear has no attention), so no competitiveness claim is made. Efficiency-oriented peers (Informer's ProbSparse, iTransformer, TimesNet, SSM baselines) remain pending (Appendix future-work item 3).
- **Single-machine systems measurements.** Forward latency (batch and lookback scaling), MACs, memory, and training time are measured on the development machine (Table 6, Figure 7 and 5). Cross-hardware results (Linux x86, GPU) and depth/width ablations are conscious scope exclusions, deferred to future work. The energy projections in Section 4.8 are derived from these counts under a stated first-order model; they are not energy measurements, and their validation against measured latency is directional.
- **Non-IID samples.** Stride-1 windows overlap heavily; samples are not independent, which standard per-window significance tests would assume. Our paired tests are at the seed level (five independent runs), where the overlap does not invalidate them; block-bootstrap or HAC-robust corrections are deferred until per-window statistics are used (Section 4.6).
- **Fixed budget, single checkpoint policy.** 10 epochs, epoch-10 weights; no early stopping or best-validation selection.
- **Non-learned routing.** The hard top-k selector gives the indexer no task-loss gradient (gradient audit: all indexer-exclusive norms exactly zero, both variants), so no learning signal reaches the selector and granularity effects are architectural. The audit establishes gradient isolation, not literal route constancy: index scores are computed from features produced by trainable upstream components, so the selected set can still change as those features drift over training. Route-overlap probes on fixed inputs across epochs and an explicitly trained selector (aux loss, relaxation, STE) are deferred future work.
- **Single-block backbone retained.** The study keeps the minimal one-block backbone deliberately (the backbone, not the attention module, is the accuracy bottleneck at this scale, Section 4.5); forecast cost is reported both ways (full-layer and forecast-required, Section 4.7), and a multilayer backbone remains future work. A transfer run grafting the four cells onto the published PatchTST backbone (ETTh1/ETTh2, five seeds each) finds all four cells tied within noise (effects $\sim$0.003, an order of magnitude below the minimal backbone). That transfer tokenizes the series into 31 patches, so the coarse-router cells can select every candidate index token; the tie therefore bounds the transfer test from below rather than showing that routing effects vanish on stronger backbones (Appendix C).
- **CPU-only, fp32, small models.** Dense-with-masking implementation choices are CPU-oriented and may not transfer to GPU-flash-attention regimes.

5.4 What this paper does not claim

Consistent with the project's pre-registered warnings: this is not a novel sparse-attention mechanism; it is not a state-of-the-art forecasting system; the word "equivalence" is not used for any accuracy comparison; no subquadratic claim is made beyond the structural column counts above; and no component is claimed to be original. The contribution is the controlled study, its measurements, and its honest boundaries.

6 Conclusion

We presented a controlled 2×2 factorial design study of compressed-and-indexed attention for long-horizon time-series forecasting, varying the KV compression rate c_{KV} and the routing resolution c_R independently under a frozen task, an auxiliary-identical contract, and parameter equalization, across 25 paired seeds. On ETTh1 direct-96 forecasting, every compressed configuration outperformed full causal attention (mean validation deltas -18.8% to -27.0%; 95/100 paired seed comparisons; Holm-adjusted $p \leq 3.3 \times 10^{-6}$; wiring-audited baseline), and the factorial separates the two axes: coarser routing deteriorates accuracy while coarser KV compression is mildly beneficial, with the pre-registered decoupling hypothesis supported on per-seed paired contrasts (exact Wilcoxon $p = 0.0074$) and its interaction localized to the $(4, 8)$ cell (Section 4.2); DRRA $(8, 4)$ cuts the attention matrix $5.22\times$ (256 vs. 49 complete columns) at equal parameters (its edge over the equal-rate cell is not significant on validation, exact Wilcoxon $p = 0.62$). Replicating the three-way comparison across twelve datasets shows the advantage is *signal-dependent*, not universal: DRRA $(8, 4)$ and CSA $(4, 4)$ beat full attention on mean validation MSE in 8/12 datasets each, with the wins concentrated on the ETT family, solar, and electricity, and slight losses on the mechanistic synthetic probes, exchange-rate validation, and weather test error. We deliberately stop short of generalization or latency claims: the factorial conclusions are established on one task, the cross-dataset pattern is descriptive and dataset-characterized, not asserted as a design law. The full four-cell replication on three contrasting datasets did not reproduce the ETTh1 sign structure, exploratory (16,4) and (32,4) probes stay flat-to-beneficial out to 32:1 outside the confirmatory tables, and a PatchTST-backbone transfer ties all four cells within noise, so rate effects are backbone-relative (that transfer's 31-token tokenization leaves the coarse-router cells selecting every candidate, so it bounds the claim from below). The indexer receives no task gradient, so granularity effects are architectural.

These results also carry a deployment-facing reading. DRRA $(8, 4)$'s structural savings—$5.22\times$ fewer attention columns and 20% fewer forward MACs than full attention at equal or fewer parameters (projection-matched, 7.8%; the dense baseline's rank-128 output projection accounts for two-thirds of the whole-model gap), and matching that projection narrows the arithmetic gap but not the accuracy gap: the retrained matched baseline scores 0.79 validation MSE, better than the as-run Normal, and every compressed cell still beats it (Section 4.6). So per-forecast arithmetic work is roughly a fifth smaller on the frozen task at batched operating points (at batch 1 the dense baseline remains fastest, Section 4.7, so this reading does not extend to single-window edge inference); where attention dominates the arithmetic, energy tracks it to a first order (Patterson et al., 2021; Wu et al., 2022), subject to the MACs-are-not-energy caveat in Section 4.7; on hardware where the attention term dominates, the gap only widens as the lookback grows (Figure 5). Two structural facts make the aggregate stakes larger than any single measurement: forecasting services run continuously at sustained query volume, so per-forecast arithmetic savings compound with volume rather than accruing once; and because the saving is architectural, it is inherited by any deployment that adopts the same design point, independent of software stack. We do not claim that translation here: the measured latency advantage is directional and single-machine (both the full-layer and the forecast-required paths are reported, Figure 6), and cross-hardware transfer remains open. But the efficiency gradient this paper maps is the one that matters operationally: content density can be coarsened without directionally costing accuracy on the smooth, redundant series that dominate production forecasting (Section 4.4).

The scaling forecast in Figure 8 extends this reading beyond the measured range. Under the audited component accounting, DRRA $(8, 4)$ removes 74.7% of attention-path work at a 32k-token context—rising to a 42.7–72.9% reduction of whole-layer work at widths from 4096 down to 128—and the closed form $S_\infty = 1 - (1/c_{KV} + 1/(2c_R))$ shows why the two axes are not interchangeable: at DRRA's settings the asymptote splits evenly between retained KV columns (12.5 points) and index scoring (12.5 points), so a further step on *either* axis (c_{KV} $8\to16$ or c_R $4\to8$) buys the same 6.2 points. That is the work-saving form of the decoupling result: the two axes are equally powerful levers on cost, but only coarsening KV is (mildly) free in accuracy on this task, so the decoupled corner takes the free lever and keeps the costly one fine. The projection is deliberately coefficient-free—it is a ratio of arithmetic counts—and it is a forecast, not a measurement: realized energy savings will be smaller wherever memory traffic or fixed costs dominate, as the batch-1 measurements already show.

The immediate roadmap is temporal-dependence-aware statistics for the cross-dataset suite, efficiency-oriented peer baselines, and hardware latency benchmarking; each is itemized in the appendix future-work list (Appendix F).

Reproducibility Statement

Every number in this paper traces to a machine-readable artifact in the repository: the frozen task specification, the pre-registered factorial design, per-run JSONL manifests for all 125 factorial/baseline runs (including the corrected-mapping re-runs of three cells), the 180-run cross-dataset suite, the external reference-point runs, the systems benchmarks, and the causality/wiring-audit test suites (Appendix E). Figures and tables are regenerated by scripts that re-read the same manifests and print verification lines against the audited summaries. The seed lists for the power extension were fixed in its design document before any extension run; the merge with the pre-registered seeds was planned at that time (Section 4.6).

Ethics Statement

This study uses public benchmark time-series datasets (cited in Section 3.1) and deterministic synthetic generators; it involves no human subjects and no sensitive data. The work is motivated by the energy cost of attention at deployment scale (Section 1): by mapping which compression axis can be coarsened without accuracy loss, it points toward cheaper long-horizon forecasting on power-constrained hardware. That motivation is quantitative but indirect—the paper reports single-machine latency and arithmetic counts, together with a first-order projection of the energy consequences of the measured work reductions under stated assumptions, rather than energy measurements, and it makes no claim about specific production systems. All findings are scoped to the frozen task and CPU scale (Section 5.3).

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Table 9: PatchTST-backbone transfer: validation MSE means over five seeds (cells are (c_{KV}, c_R) pairs; lower is better). All four cells tie within noise on both datasets; forward latency (B32 = batch of 32, mean) 37.4 $(4,4)$ / 33.7 $(8,4)$ / 33.0 $(4,8)$ / 29.4 $(8,8)$ ms, i.e. $(8,4)$ is 1.11× and $(8,8)$ 1.27× the speed of $(4,4)$ at tied accuracy (effects $\sim$0.003, an order of magnitude below the minimal backbone).

Dataset	$(4,4)$	$(8,4)$	$(4,8)$	$(8,8)$
ETTh1	0.3730	0.3705	0.3745	0.3704
ETTh2	0.2559	0.2613	0.2614	0.2613
Forward ms (B32 mean)	37.4	33.7	33.0	29.4

A Documented negative results

Six failed designs were tried and documented during the development of this family (each with runnable or archived implementations and failure modes): sparse-gather routing ($\sim$5× slower than dense masking on CPU); a hierarchical (multi-scale pyramid) indexer (consistently worse than the flat indexer); a multi-scale KV pyramid at 224 KV tokens (overparameterized for the budget); a learnable block-bias offset (NaN instability in $\sim$50% of runs; root cause later traced to uninitialized memory—'torch.empty' biases—and fixed); a linear-pool indexer with gated residual KV (loses the per-head × per-dimension routing flexibility of the softmax-overlap gate); and stacked 2:1 → 2:1 compression (harder to optimize within the budget). Negative results of this kind are a deliberate part of the reproducibility package; they are secondary to the paper's core claim.

B Surrogate attention (separate line of work)

Distilling the attention block into a cheap inference-time student is a separate line of work, deferred to a separate publication: it is *not* part of this paper's claims, and none of its numbers are apples-to-apples with the frozen-task results in Section 4 (they predate the frozen protocol). This pointer exists only to acknowledge the overlap with attention-distillation work (Goldstein et al., 2025) and to record the retention decision: the surrogate line, including the equal-budget small-transformer control and distillation-epoch alignment, lives outside this manuscript.

C Exploratory rate sweep

Two post-hoc cells, $(16,4)$ and $(32,4)$ (25 ETTh1 seeds each, same protocol), probe how far KV compression goes while routing stays fine. They are kept out of the confirmatory tables and figures by design; Figure 11 shows them together with the factorial cells, marked with an asterisk. $(16,4)$ reaches the best mean (0.5810 validation, 0.8100 test) and $(32,4)$ lands inside the top group (0.6122 / 0.8586): the KV-rate profile stays flat-to-beneficial out to 32:1 on ETTh1, and 4 KV blocks is the sensible floor where this study stops. Neither probe is statistically separated from the named dual-rate corner in paired 25-seed contrasts: for $(16,4)$ versus $(8,4)$ the paired validation difference is -0.038 (uncorrected exact Wilcoxon $p = 0.08$, 16 of 25 seeds favoring $(16,4)$) and the test difference is -0.034 ($p = 0.06$, 16 of 25); for $(32,4)$ the corresponding differences are -0.007 ($p = 0.51$) and $+0.015$ ($p = 0.41$). The probes therefore remain exploratory: they motivate the long-context forecast of Section 4.8, but they enter no confirmatory claim, and promoting either to the named design corner would require independent replication (fresh seeds and the cross-dataset suite, as itemized in Appendix F).

The same four cells were also grafted onto the published PatchTST backbone (ETTh1/ETTh2, five seeds each) as a transfer test. Table 9 shows all four tied within noise on both datasets. The transfer is boundary-limited, and the tie does not by itself show that a strong backbone washes out routing: with lookback 256, patch length 16 and stride 8 the backbone carries 31 tokens, so at c_R=4 there are 6–7 candidate index tokens and at c_R=8 only 3 against k=4—the coarse-router cells select every candidate, and the fine-router cells select 4 of 6–7, which at c_{KV}=8 map onto at most three coarse blocks. Routing resolution is therefore barely exercised in this transfer, so the tie is consistent both with backbone-dependence of the rate effects and with exaggerated routing saturation at this token count; it bounds the transfer claim from below. Figure 10 shows the same tie on the accuracy axis with separation on the speed axis.

D Energy projection: derivation, sensitivity, and audit

This appendix documents the projection used in Section 4.8: the operation counts it reads, its sensitivity to the model's assumptions, and its checks against the audited single-machine measurements. The purpose is to make every projected

Table 10: Forward component accounts at L=256, per sequence, in millions of multiply–accumulate operations (released accounting). The analytic totals equal the released totals exactly; the figure script asserts this equality before drawing Figure 8. Dashes mark components a configuration does not have.

Component	Normal	$(4, 4)$	$(8, 4)$
Wrapper (input projection, MLP, horizon head)	33.87	33.87	33.87
Query low-rank projection	4.19	4.19	4.19
Key/value projection (tied)	1.05	1.05	1.05
Attention core (scores and values)	16.78	5.31	3.21
Router	—	2.10	2.10
Compressors (two-series)	—	6.29	6.29
Output projection	20.97	10.49	10.49
Total	76.86	63.30	61.20

number in the main text traceable to a released artifact and to state plainly what the projection does and does not support.

Operation counts. Table 10 lists the per-component forward arithmetic of the three named configurations at L=256, in millions of multiply–accumulate operations per sequence. These are the released component accounts: the analytic totals in the last row equal the audited totals exactly (76.86M, 63.30M, and 61.20M for Normal, CSA $(4, 4)$, and DRRA $(8, 4)$), and the figure script asserts this equality before drawing Figure 8. The router row counts what the module executes: the routing query is the main multi-head query and the index stream is single-headed, so the score matmul runs at $[B, H, S, d_h] \times [B, 1, d_h, M]$, i.e. $hd_h = d = 128$ MACs per (query, index-token) pair, as verified against the executed tensor shapes in the released accounting log. Two asymmetries between the columns are properties of the as-run configurations, not of the projection: the dense baseline uses a rank-128 output projection while the compressed cells use the equalized rank-64 bottleneck, and only the compressed cells pay the router and compressor terms. Of the 15.66M whole-model gap between Normal and DRRA $(8, 4)$, 10.49M is the output projection alone; matching that rank leaves a 7.8% whole-model difference. That matched baseline was retrained too (25 paired seeds): it improves dense accuracy over the as-run Normal and every compressed cell still beats it, so the accuracy gap is not a projection artifact. Operations excluded from the accounting (RMS normalization, rotary embeddings, mask construction, the head-mixing weight of the index scores, and the softmax normalization itself) are negligible at these dimensions and are excluded consistently across configurations.

Derivation. The derivation of Equations 2–4—each factor's origin, each term's power of L, and the form of the whole-layer ratio—is given with the equations in the main text (Section 4.8). This appendix supplies the audit layer around it: the released operation counts (Table 10), the sensitivity band, the validation checks, and the scope statement.

Sensitivity. Three perturbations bound the assumptions. (i) *Auxiliary weighting.* Reweighting the router and compressor terms by a common factor λ leaves the 32k DRRA attention-path saving at 81.1% (λ=0.5), 74.7% (λ=1), and 61.9% (λ=2): the direction is unchanged, and the spread sets how much of the result rests on how the two auxiliary terms are costed. (ii) *Single-head indexer.* The executed router scores each index token with every head of the shared query, at $hd_h = d$ MACs per comparison, so the frozen h=4 geometry pays four times what a single-head indexer would; a single-head indexer (as in a Lightning-Indexer-style design, which this module does not implement) would raise the 32k DRRA projection from 74.7% to 84.0% and the asymptote from 75.0% to 84.4%. The executed arrangement is therefore the conservative one for the saving claim, and the asymptotic router share $1/(2c_R)$ does not shrink with the head count at fixed d. (iii) *Memory traffic.* The key/value traffic of the attention core scales with the same selected-column count as its arithmetic, so an additive energy cost per read column changes the effective coefficient but not the column ratio; the structurally distinct terms are the router (quadratic in L, with constant $1/c_R$) and the compressors (linear in L), which the auxiliary-weighting band covers. Excluded from the model throughout: memory stalls, frequency scaling under load, leakage, and idle power. A complete energy evaluation would need per-level memory counters and platform power traces; we report none and claim none.

Table 11: Verification-audit outcomes (all logs released with the source). LAST = forecast-required path; FULL = full-layer forward; B32 = batch of 32; paired deltas are within-round.

Audit	Result	Headline numbers
Last-query equivalence	PASS (max diff 0.00)	Earlier outputs provably unused
Forecast-required latency (B32)	PASS (bit-exact $\sim 5 \times 10^{-7}$)	Normal 2.6–3.9 ms; compressed 5.8–9.5 ms
Paired full-layer deltas	Directional	$(8, 4)$ − Normal -9.0 ± 1.7 ms/round
Unique routed blocks/query	Measured	$\approx$14 of 64 (equal-rate); $\approx$10 of 32 (dual-rate)
Router gradient audit	No task gradient	All indexer grads exactly 0, both variants; route membership probed: mean Jaccard 0.83 epoch 1 vs 10, set size $\approx$10 of 32 flat
Projection-matched baseline	Advantage survives	Matched-rank dense retrain (25 seeds): beats the as-run Normal, loses to every compressed cell (paired val deltas -0.110 to -0.178, intervals exclude 0)
Mechanism probes (regenerated)	Current implementation	Normal/$(8, 4)$ bit-identical; $(4, 4)$ updated post-fix
Causality probes (masking/ties)	PASS, all six rate pairs	Past output/sets bitwise unchanged under adversarial future noise
Index-to-block mapping	Fixed + re-run	Pre-fix: 3 of 4 cells misrouted; post-fix verified
Component accounting	Rechecked	Router counted per executed tensor shapes (hd_h per comparison); totals 76.86/63.30/61.20M
Factorial recompute (25 seeds)	MATCH	All headline values; H1 exact $p = 0.0074$
Backbone transfer (PatchTST)	Tie within noise	All four cells tied; speed $(8, 4)$ 1.11x, $(8, 8)$ 1.27x vs $(4, 4)$
Exploratory speeds (FULL)	Plateau past 16:1	1.18x $(8, 4)$, 1.22x $(16, 4)$/$(32, 4)$ vs Normal
Scaling forecast	Exact + directional	Totals match the released accounts; latency rank $\rho = 1.000$

Validation. Three checks are available from the released measurements. First, exact arithmetic: the analytic totals reproduce the audited component accounts at L=256 exactly, as asserted by the figure script. Second, rank agreement with time: across the audited lookback sweep, measured forward latency ranks with analytic attention-path work at Spearman $\rho = 1.000$ (batch 32, n=15) and $\rho = 0.992$ (batch 1, n=18), and the paired DRRA-versus-Normal saving ratio correlates with the work ratio at Pearson $r = 0.96$ (n=5, batch 32). Third, magnitude agreement where commensurate: the whole-layer projection for the study's backbone is 50.5% at L=2048/batch 32 against 56.2% measured, and 60.8% at L=4096/batch 1 against 63.3% measured. The measured saving exceeds the arithmetic projection at both points, consistently with the dense baseline being memory-bound at these contexts (its materialized score tensor dominates its time), so wall-clock cost grows faster than MAC count there; the work model excludes memory traffic by construction.

Scope of the claim. The projection answers "how much arithmetic does the design remove," not "how many joules does it save." Under the stated first-order model the answer is hardware-independent; on any real hardware the realized saving will be smaller wherever memory or fixed costs dominate, as the batch-1 measurements already show. No claim is made about GPUs, accelerators, or production systems.

E Reproducibility

Every number in this paper traces to a repository artifact: the frozen task specification (the sole authority on task definition), the pre-registered factorial design, the run harness and its machine-readable results manifests (125 factorial/baseline runs at n=25 + 180 cross-dataset runs + external reference-point runs, JSONL), a verified per-module parameter registry, causality-verification and wiring-audit suites (all six variants pass future-invariance and exact-zero-future-gradient checks at the frozen geometry; the per-cell mapping regression extends the future-safety guard to every tested rate pair and to non-divisible lengths, and dedicated index-score-masking and tie-breaking probes now run at every tested rate pair (adversarial future-noise masking and tie duplication leave past outputs, selected token sets and mapped block sets bitwise unchanged; in the non-divisible 31-token regime the fan-out cell emits verified adjacent block pairs with per-block safety); the Normal baseline is bit-identical to a naive causal reference), and a sanity-test suite covering shapes, gradients, parameter counts, and tiny-task fitting (including a per-cell index-to-block mapping regression test). Supplementary audits ship as machine-readable logs alongside the manifests: last-query equivalence, the router gradient audit, canonical last-query latency with paired deltas, the index-to-block mapping audit (with the pre-fix misrouting finding and the re-run of the affected cells), the component-accounting recheck (executed-matmul shapes and corrected per-component counts, which is the source of the totals in Table 10), the paired per-seed conditional effects of the factorial with their intervals, and the unrounded 25-seed factorial recomputation. Table 11 collects the audit outcomes in one place.

Reported means and deltas are computed from the audited manifests; the figure-generation scripts in this paper's source tree read the same manifests, so every figure is regenerable. The scaling forecast is regenerated in the same way and asserts exact equality with the released component accounts before drawing. The freeze-clean package — environment lockfile, dataset SHA-256 manifests, automated figure/table regeneration scripts, and runnable test suites (causality, gradient masks, wiring audit, data integrity) — is all in the repository. The source tree is under git with a freeze baseline (commit recorded in the project changelog); the remaining reproducibility steps at release time are a tagged archive with a DOI (e.g., Zenodo) and the official venue style swap.

F Future work

Scope completed in this study versus open follow-ups, by axis:

1. **Statistics.** Factorial-level inference is complete: $n{=}25$ paired seeds; paired Wilcoxon with Holm correction over cell-vs-Normal contrasts (Section 4.6); per-seed main-effect contrast for H1. Deferred with justification: mixed-effects model (the 2×2 already identifies the three fixed effects; deferred as a robustness re-expression), HAC/block bootstrap (seed-level analyses), TOST/ROPE (no equivalence claims); temporal-dependence-aware statistics for the cross-dataset suite remain future work. Realized-budget control at matched unique blocks, and any test-primary/MAE re-headline, are deferred by design (reported as measured, with test alongside validation; per-seed values in the released manifests).

2. **Datasets.** The suite covers the ETT family (ETTh1, ETTh2, ETTm1, ETTm2), large multivariate (Electricity, Traffic, Weather), different dynamics (Exchange, Solar-Energy), and three mechanistic synthetic families (sinusoids, change points, delayed-copy), all under the frozen protocol with 5 paired seeds (180 runs; per-seed tables in the released results manifests). Remaining: horizon ($H \in \{192, 336, 720\}$) and lookback ($L \in \{512, 720, 1024\}$) sweeps. Copy-vs-noise isolation with fresh random content, one-factor smoothness/noise/localization sweeps with a selection-guidance figure, and independent confirmatory replication of the exploratory (16,4)/(32,4) probes (fresh seeds plus cross-dataset) are deferred to follow-up work.

3. **Baselines.** DLinear (Zeng et al., 2023) and PatchTST (Nie et al., 2023) under the identical frozen protocol as external reference points, not peer methods (Section 4.5): DLinear on all twelve datasets; PatchTST on the eight datasets where its channel-independent training fits the single-machine CPU budget (declared omissions on solar/electricity/traffic/weather). Remaining as future work: iTransformer, Informer, TimesNet, and an SSM baseline. Diagnostic attention controls at matched budget (all-blocks, window-only, fixed/random routing) are likewise deferred.

4. **Hardware/latency.** Development-machine measurement is complete: Mean/p90/p99 forward latency with bootstrap CIs, throughput, analytic MACs, peak-memory accounting, training time, and batch/lookback scaling sweeps (batch 1–256, $L \in \{128, \ldots, 4096\}$); see Section 4.7. Remaining as future work: cross-hardware transfer (x86 CPU, GPU) and depth/width ablations — conscious scope exclusions of this study.

5. **Literature re-audit:** the 2026-08-16 sweep of the 2024–2026 routing/compression-attention literature found no counterexample to the *to our knowledge* claim (Section 2); a fresh sweep before locking the title remains good practice.

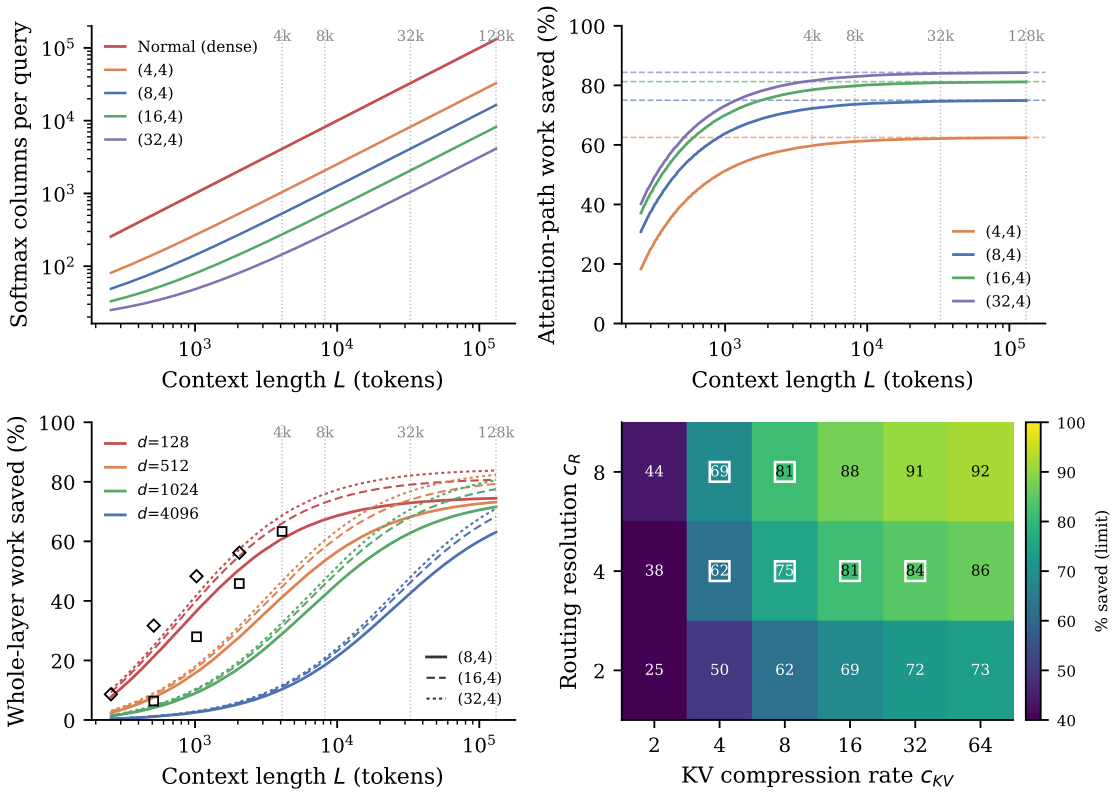


Figure 8: Structural scaling forecast for the compressed-and-indexed family, computed from the audited component accounting with no new runs. **(a)** Softmax columns per query (L = context length): dense attention grows with L, compressed cells with $L/c_{KV} + W + 1$, where W=16 is the fixed sliding window and the $+1$ is the always-visible sink column. **(b)** Attention-path work saved relative to dense attention; dashed lines are the closed-form asymptotes S_∞ (the $L \to \infty$ limit), and savings saturate once the quadratic score and router terms dominate the linear compressor cost. **(c)** Whole-layer work saved for backbones of increasing width (standard $12d^2$ non-attention cost per token per layer: $4d^2$ q/k/v/o projections plus $8d^2$ MLP): $(8, 4)$ solid, $(16, 4)$ dashed, and $(32, 4)$ dotted, colored by width, with the measured end-to-end $(8, 4)$ time savings for the study's backbone overlaid as open markers (squares: batch 1; diamonds: batch 32; positive savings only). The measured points sit on or below the projection, as expected, because they include non-attention work. **(d)** Closed-form asymptotic savings over the design plane; white squares mark the tested cells. All curves are analytic projections, not measurements; the accompanying script asserts exact equality with the released component accounts at L=256 before drawing.

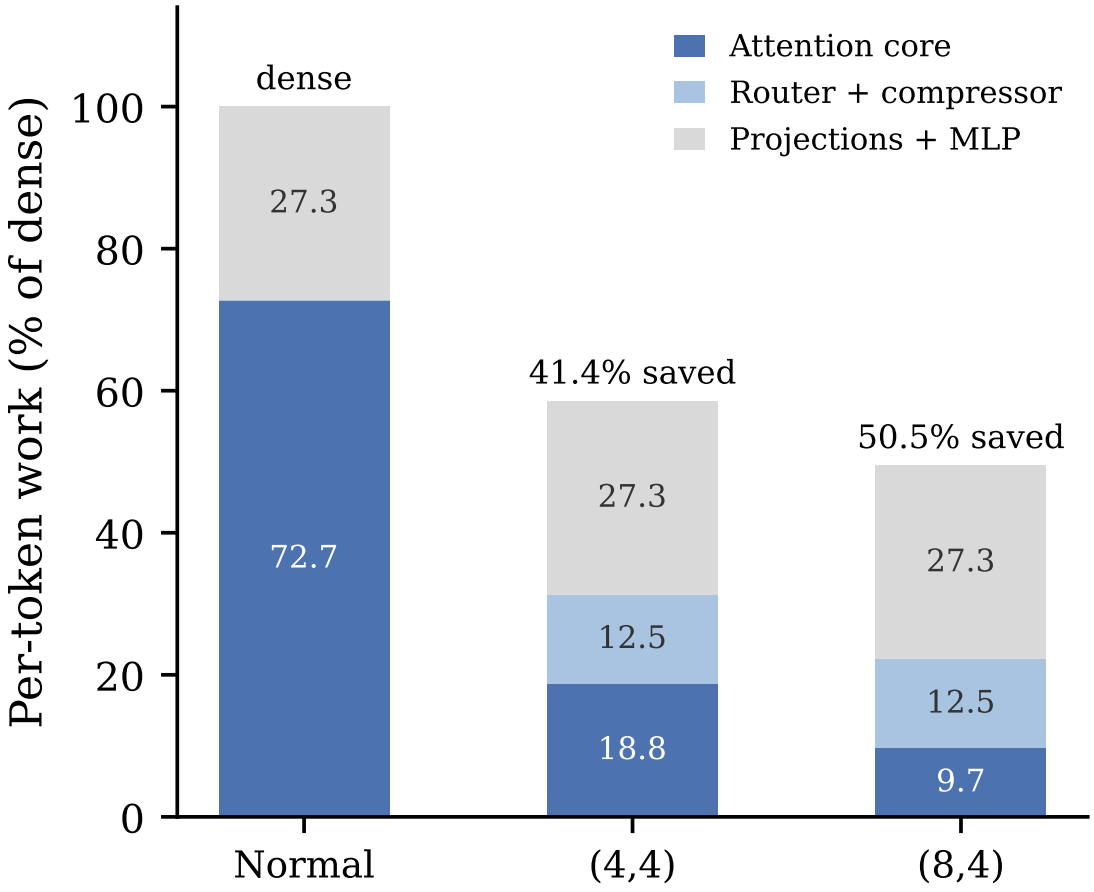


Figure 9: From work to energy at the study operating point (L=2048 = context length, d=128 = hidden size). Under the single-coefficient model (energy E proportional to total arithmetic work) the per-token work shares are also its energy shares, so the bars are normalized to the dense model and imply no absolute energy. The attention path alone saves 57.0% $(4, 4)$ and 69.5% $(8, 4)$ of attention work (Equation 3), but the fixed per-token block of $12d^2$ projections-and-MLP operations ($4d^2$ q/k/v/o projections plus $8d^2$ MLP) (Equation 4) is untouched by the cell definition and caps the whole-model saving at 41.4% and 50.5%. Counted from the released component accounts (Table 10); no new runs.

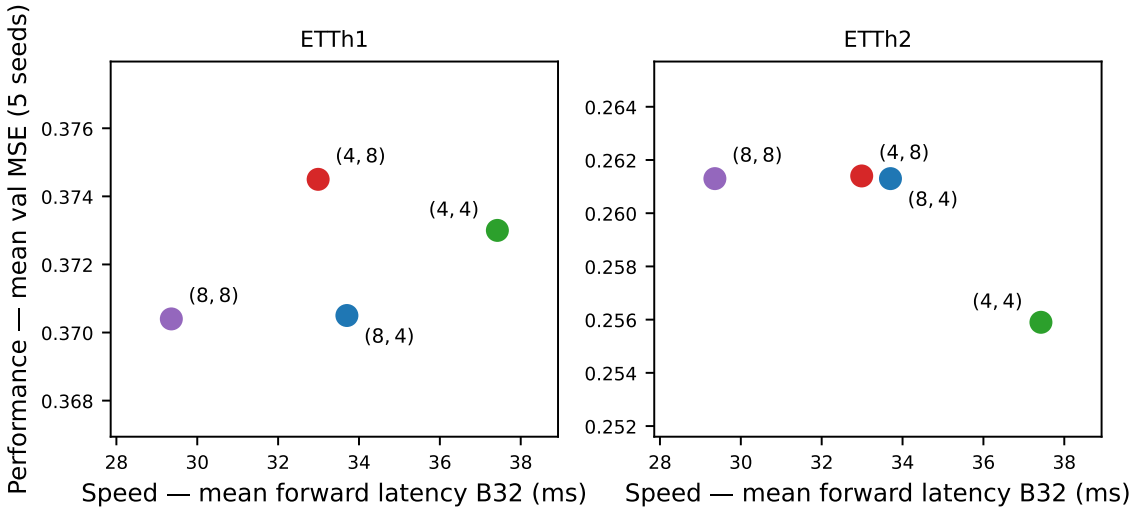


Figure 10: PatchTST-backbone transfer, performance vs. speed: y-axis = mean validation MSE (mean squared error) over five seeds (lower is better); x-axis = mean forward-pass latency at batch 32 (lower = faster = further left). Point labels are the cells' (c_{KV}, c_R) rates. Left ETTh1, right ETTh2. Accuracy ties within noise on both sets while latency separates the cells ($(8,4)$ is $1.11\times$ and $(8,8)$ $1.27\times$ the speed of $(4,4)$ at tied accuracy). Points are means only with no error bars; single machine, descriptive; FULL path only (the PatchTST head reads all patches).

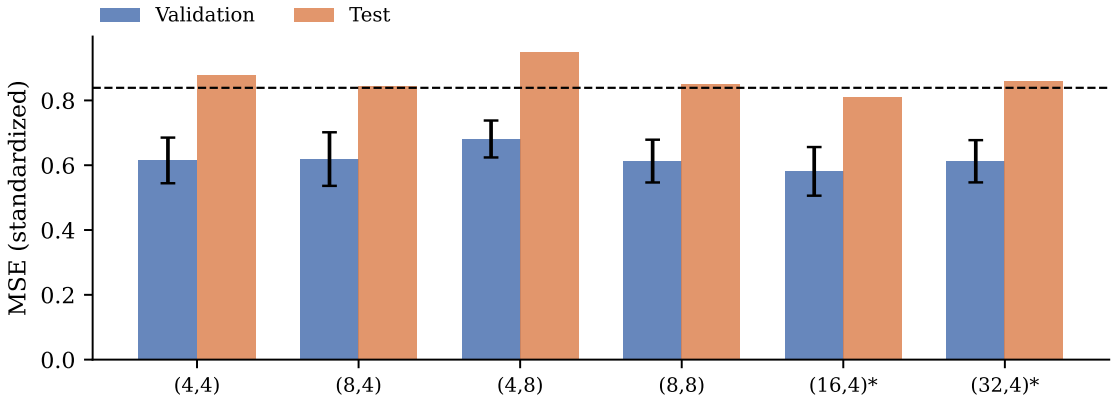


Figure 11: ETTh1 validation (blue) and test (orange) means $\pm$ one seed standard deviation over 25 seeds for all cells (cells are (c_{KV}, c_R) pairs; lower is better). Starred cells (labels with $*$) are post-hoc exploratory probes, excluded from the confirmatory tables. The horizontal dashed line marks the full-attention (Normal) validation baseline at 0.8389 MSE, so validation bars below it beat full attention; test values lie higher and are not marked.